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MIL-M-28787/523
21 APRIL 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLY/MATH PACKAGE MODULE, KEY CODE DWAE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiply/math package module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 6.063 failures per million operating hours for class I or 15.581 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DWAE.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A9	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B9	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc Stress	IV	S	--------The module shall pass all remaining tests------						
D1-D4	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D5,D6	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D7	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D8	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D9	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D10	Ibd	V	S	0.0	8.4	0.0	8.4	0.0	8.4	mA
D11-D17	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D19,D20	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D21-D37	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E3	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E4-E9	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E10-E72	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E73-E75	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E76	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E77	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E78	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E79	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E80	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E81	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E82	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E83	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E84-E99	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
F1-F3	Iih	VII	X	-0.55	0.07	-0.55	0.07	-0.55	0.07	mA
F4-F6	Iih	VII	X	-0.4	0.08	-0.4	0.08	-0.4	0.08	mA
F7	Iih	VII	X	-1.0	0.10	-1.0	0.10	-1.0	0.10	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F8-F10	Ith	VII	X	-0.54	0.06	-0.54	0.06	-0.54	0.06	mA
F11	Ith	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F12	Ith	VII	X	-4.6	0.12	-4.6	0.12	-4.6	0.12	mA
F13	Ith	VII	X	-5.2	0.14	-5.2	0.14	-5.2	0.14	mA
F14	Ith	VII	X	-4.6	0.12	-4.6	0.12	-4.6	0.12	mA
F15	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F16	Ith	VII	X	-17.6	0.48	-17.6	0.48	-17.6	0.48	mA
F17-F21	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F22-F26	Ith	VII	X	-0.54	0.06	-0.54	0.06	-0.54	0.06	mA
F27	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F28-F43	Ith	VII	X	-0.22	0.04	-0.22	0.04	-0.22	0.04	mA
F44	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F45	Ith	VII	X	-4.85	0.08	-4.85	0.08	-4.85	0.08	mA
F46,F47	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F48-F64	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G1-G3	Ith	VIII	X	-0.55	0.07	-0.55	0.07	-0.55	0.07	mA
G4-G6	Ith	VIII	X	-0.4	0.08	-0.4	0.08	-0.4	0.08	mA
G7	Ith	VIII	X	-1.0	0.10	-1.0	0.10	-1.0	0.10	mA
G8-G10	Ith	VIII	X	-0.54	0.06	-0.54	0.06	-0.54	0.06	mA
G11	Ith	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G12	Ith	VIII	X	-4.6	0.12	-4.6	0.12	-4.6	0.12	mA
G13	Ith	VIII	X	-5.2	0.14	-5.2	0.14	-5.2	0.14	mA
G14	Ith	VIII	X	-4.6	0.12	-4.6	0.12	-4.6	0.12	mA
G15	Ith	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G16	Ith	VIII	X	-17.6	0.48	-17.6	0.48	-17.6	0.48	mA
G17-G21	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G22-G26	Ith	VIII	X	-0.54	0.06	-0.54	0.06	-0.54	0.06	mA
G27	Ith	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G28-G43	Ith	VIII	X	-0.22	0.04	-0.22	0.04	-0.22	0.04	mA
G44	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G45	Ith	VIII	X	-4.85	0.08	-4.85	0.08	-4.85	0.08	mA
G46,G47	Ith	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G48-G64	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
H1-H99	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J100	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K16	Iozh	XI	X	-0.08	0.08	-0.08	0.08	-0.08	0.08	mA
K17-K32	Iozl	XI	X	-0.08	0.08	-0.08	0.08	-0.08	0.08	mA
L1-L11	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L12	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L13-L33	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M	PSFC	XIII	S	9.0	16.9	9.0	16.9	9.0	16.9	uF
N	Icc	XIV	X	600	2500	600	2500	600	2500	mA
P1,P2	Functional	XV	X	----------		See table XV		----------		
Q1,Q2	Dynamic	XVI	S	----------		See table XVI		----------		
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI		----------		
R1-R3	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R4	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R5	Ci/o	XVIII	Q	4	125	4	125	4	125	pF
R6-R8	Ci/o	XVIII	Q	4	150	4	150	4	150	pF
R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	18	228	18	228	18	228	pF
R11-R17	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R18	Ci/o	XVIII	Q	8	125	8	125	8	125	pF
R19,R20	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R21-R37	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R38	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R39-R41	Ci/o	XVIII	Q	2	175	2	175	2	175	pF
R42-R47	Ci/o	XVIII	Q	8	125	8	125	8	125	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 3°C 60 ± 3°C		Class II 4/ -55 ± 3°C 85 ± 3°C		Units
				Min	Max	Min	Max	Min	Max	
R48-R53	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R54,R55	C1/o	XVIII	Q	8	175	8	175	8	175	pF
R56	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R57	C1/o	XVIII	Q	8	175	8	175	8	175	pF
R58	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R59-R61	C1/o	XVIII	Q	8	125	8	125	8	125	pF
R62	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R63-R73	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R74,R75	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R76-R78	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R79,R80	C1/o	XVIII	Q	2	125	2	125	2	125	pF
T	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/523-1 for class I, and M28787/523-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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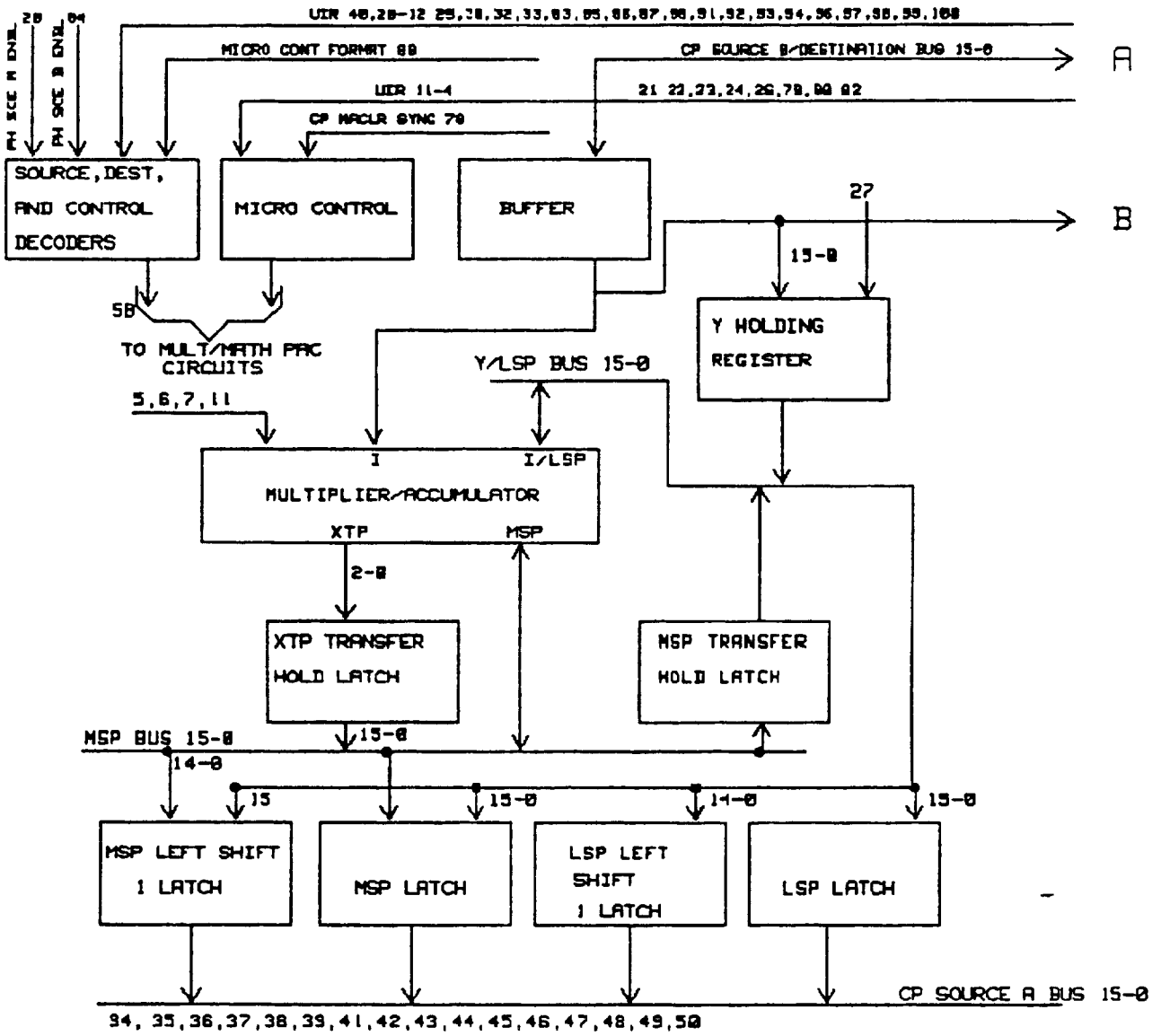


FIGURE 1. DWAE functional block diagram.

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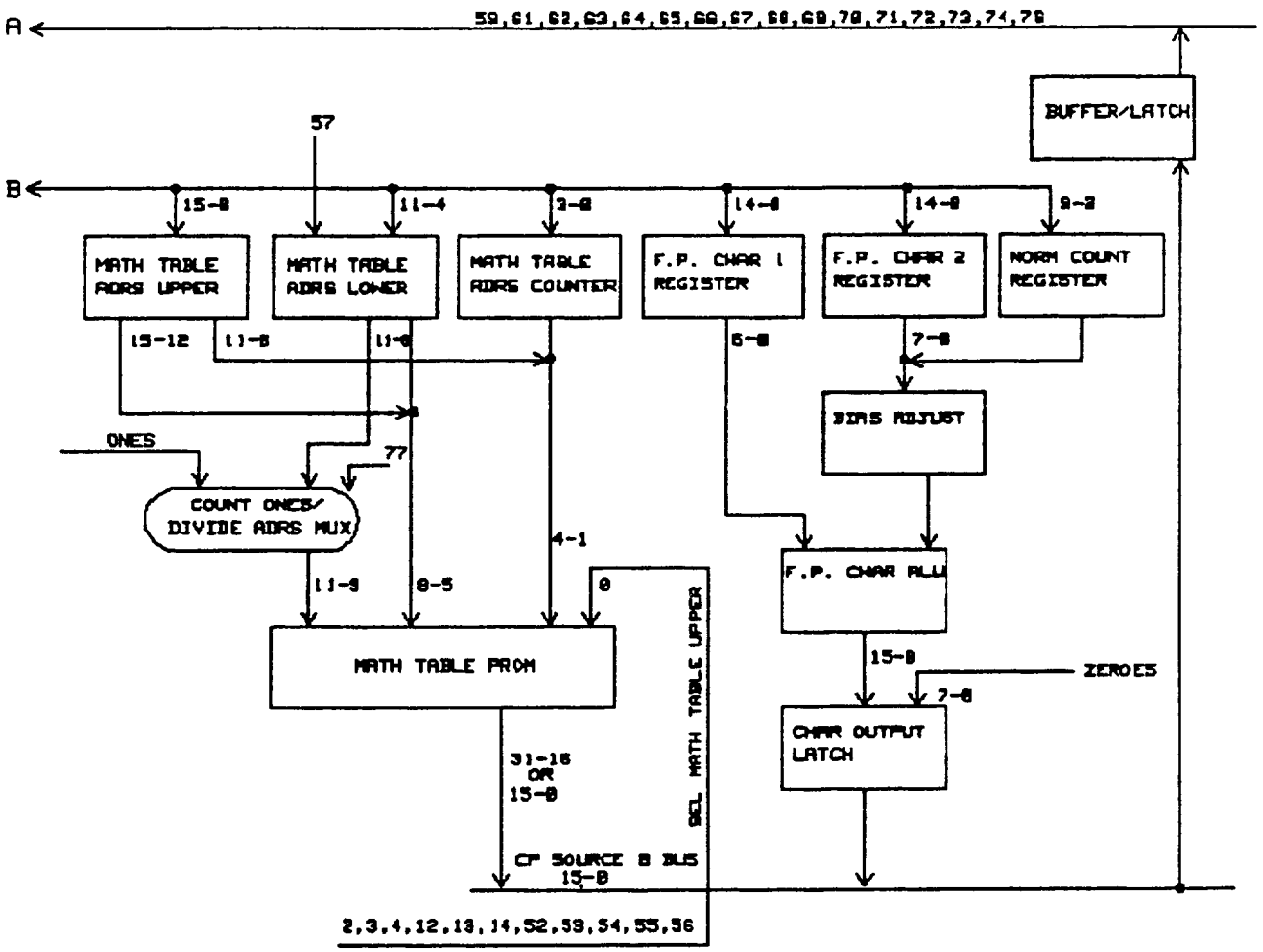


FIGURE 1. DWAE functional block diagram - Continued.

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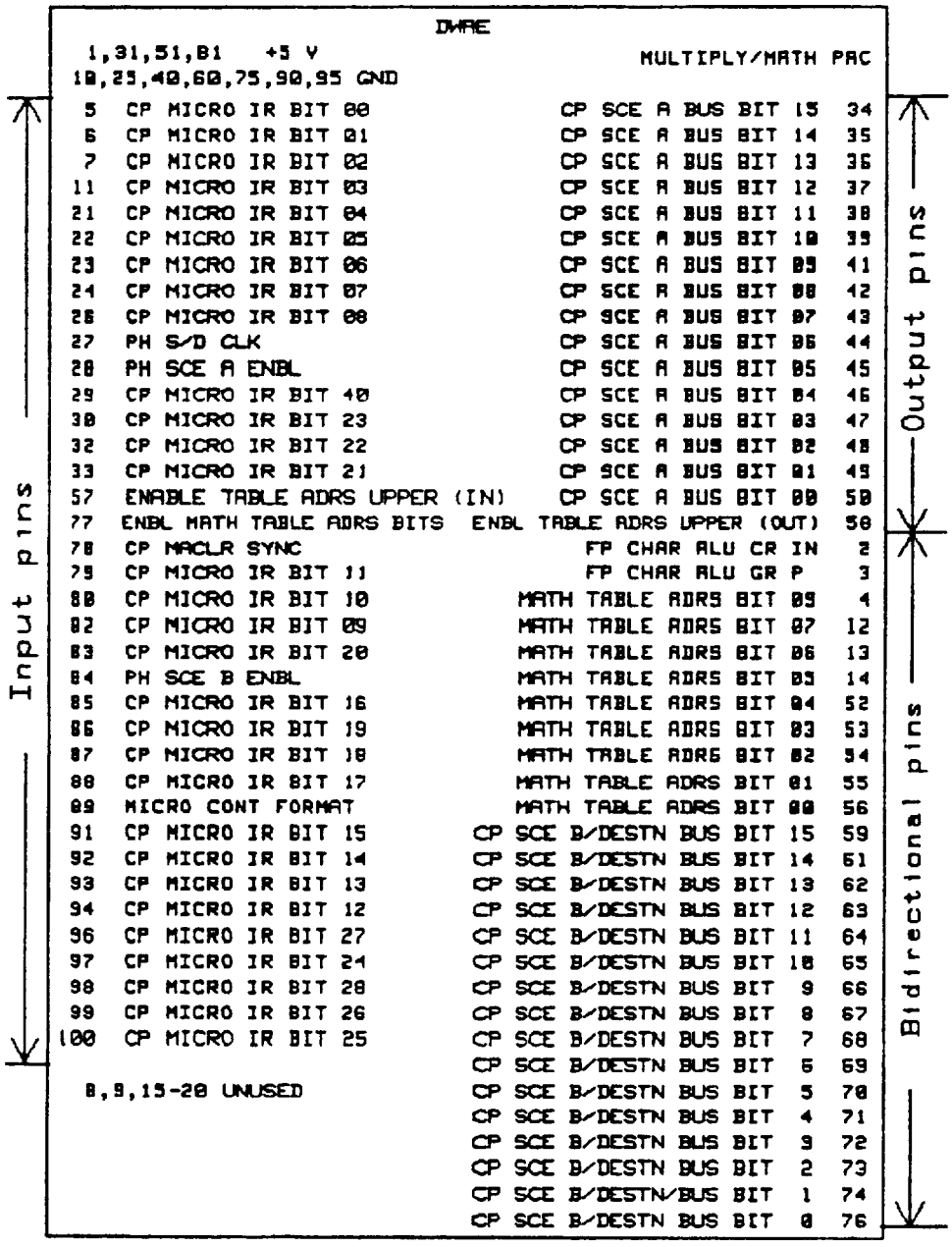


FIGURE 2. DWAE pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M

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through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

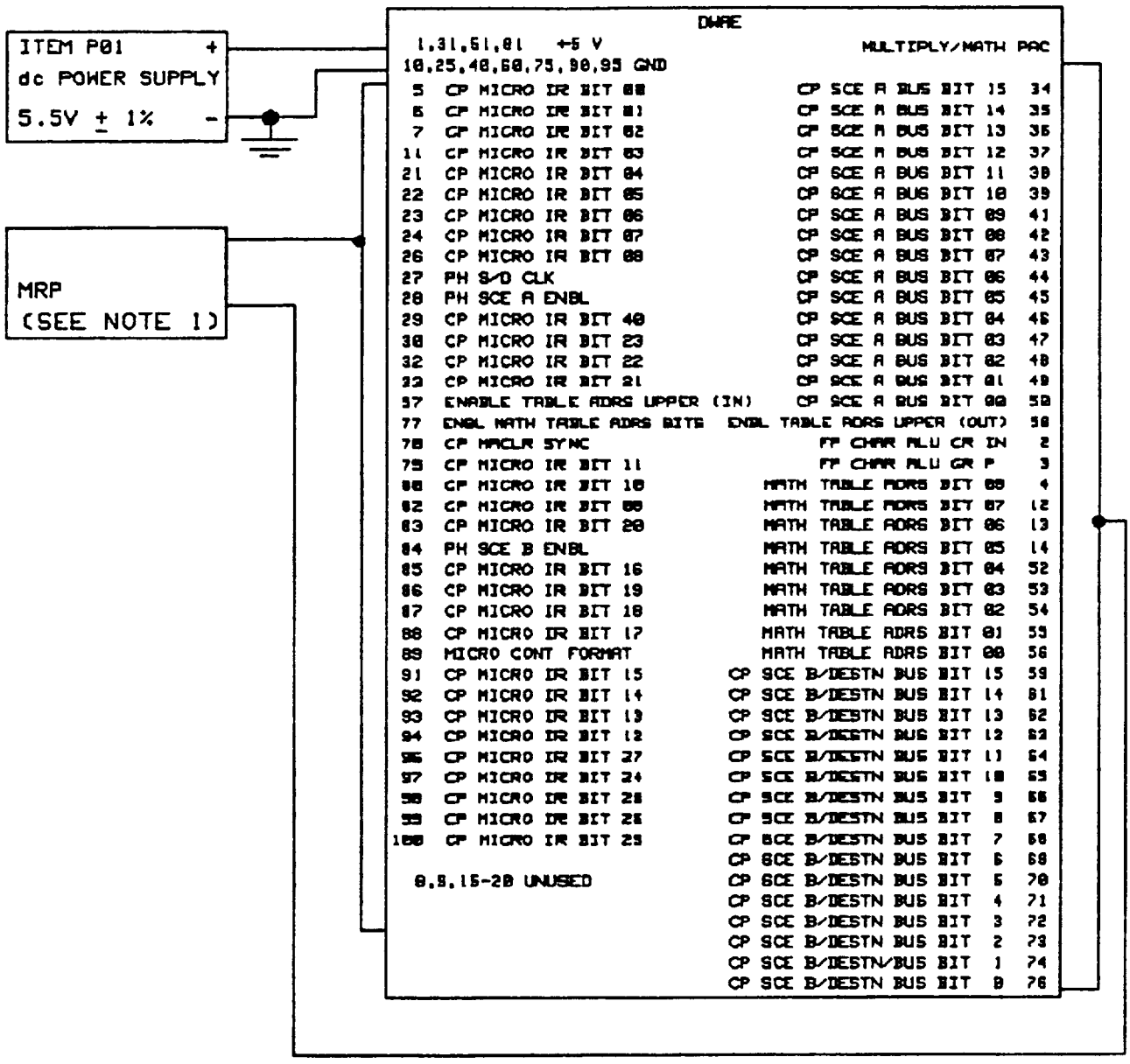
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. DWAE operating temperature and component temperature test circuit.

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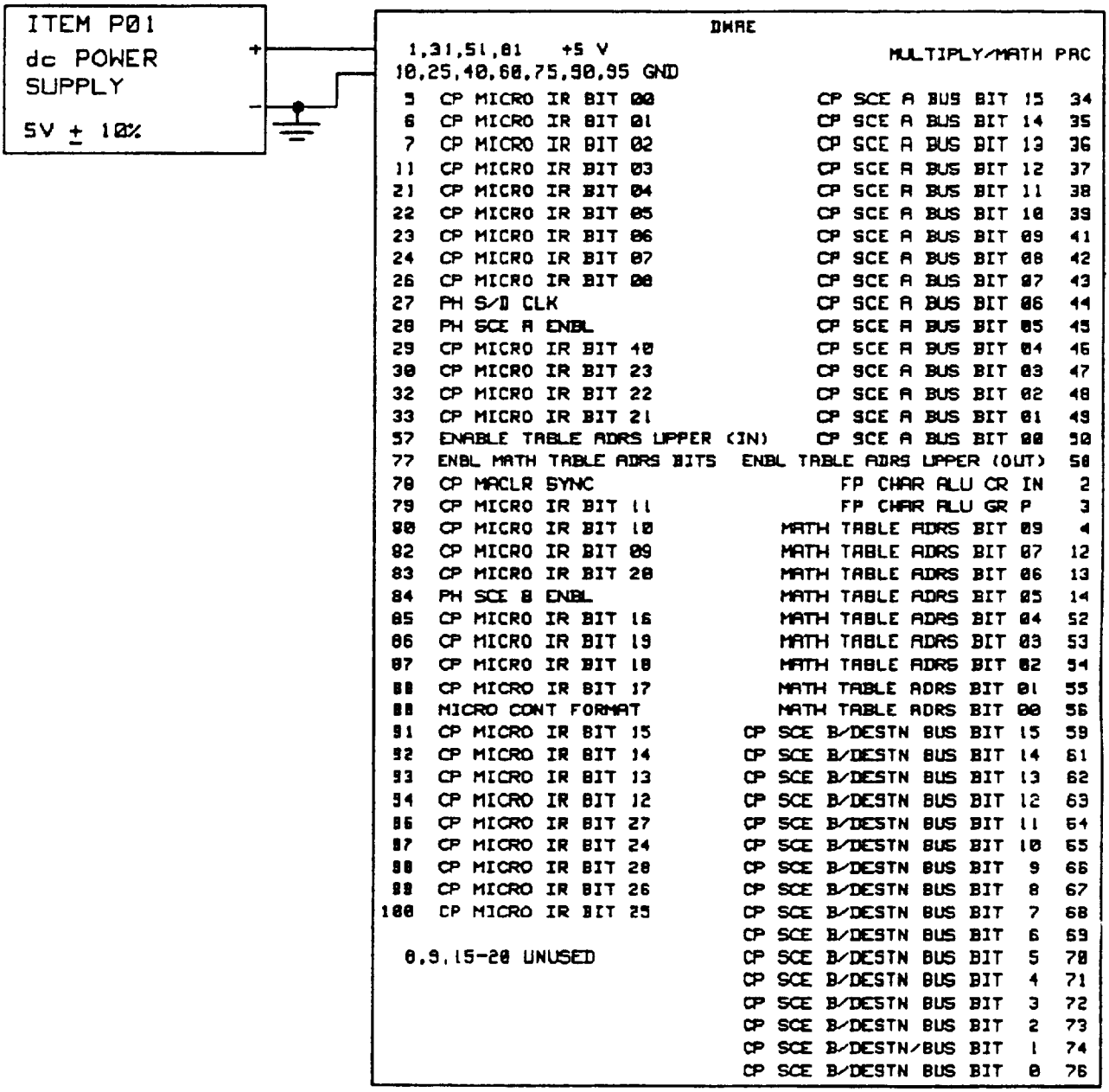
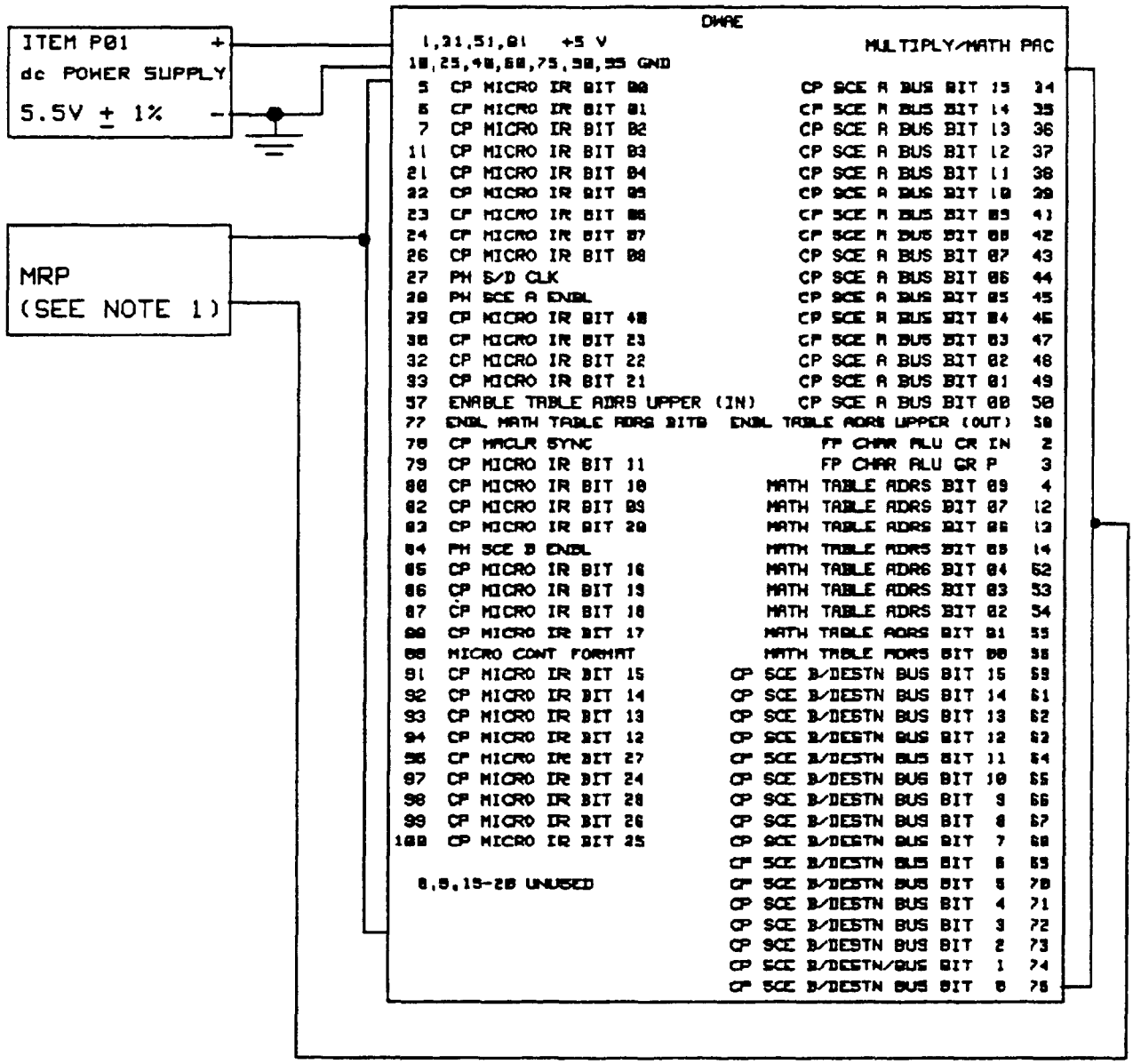


FIGURE 4. DWAE vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. DWAE life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Ci/o = input and output capacitance.
(b) Cin = input capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 13.75 W (max); 8.23 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 0	NA	0.4	2.4	-0.4	0.08	1.0	75
IR BIT 1	NA	0.4	2.4	-0.4	0.08	1.0	75
IR BIT 2	NA	0.4	2.4	-0.4	0.08	1.0	75
IR BIT 3	NA	0.4	2.4	-1.0	0.10	1.0	75
IR BIT 4	-1.2	0.4	2.4	-1.8	0.06	0.3	125
IR BIT 5	-1.2	0.4	2.4	-4.6	0.12	0.3	150
IR BIT 6	-1.2	0.4	2.4	-5.2	0.14	0.4	150
IR BIT 7	-1.2	0.4	2.4	-4.6	0.12	0.3	150
IR BIT 8	-1.2	0.4	2.4	-2.0	0.05	1.0	75
PH S/D CLK	-1.2	0.4	2.4	-17.6	0.48	8.4	228
SCE A EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 40	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 23	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 22	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 21	-1.2	0.4	2.4	-0.6	0.02	0.1	75
EN ADR UP	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EN M ADR BIT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MACLR SYNC	-1.2	0.4	2.4	-4.85	0.08	0.4	125
IR BIT 11	-1.2	0.4	2.4	-1.2	0.04	0.2	75
IR BIT 10	-1.2	0.4	2.4	-1.2	0.04	0.2	75
IR BIT 9	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 20	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SCE B EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 16	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 19	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 18	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 17	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CONT FRMT	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 15	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 14	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 13	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 12	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 27	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 24	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 28	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 26	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IR BIT 25	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ALU CR IN	NA	0.4	2.4	-0.55	0.07	NA	125
ALU GR P	NA	0.4	2.4	-0.55	0.07	NA	125
ADR BIT 8	NA	0.4	2.4	-0.55	0.07	NA	125
ADR BIT 7	NA	0.4	2.4	-0.54	0.06	NA	175
ADR BIT 6	NA	0.4	2.4	-0.54	0.06	NA	175

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ADR BIT 5	NA	0.4	2.4	-0.54	0.06	NA	175
ADR BIT 4	NA	0.4	2.4	-0.54	0.06	NA	150
ADR BIT 3	NA	0.4	2.4	-0.54	0.06	NA	125
ADR BIT 2	NA	0.4	2.4	-0.54	0.06	NA	125
ADR BIT 1	NA	0.4	2.4	-0.54	0.06	NA	125
ADR BIT 0	NA	0.4	2.4	-0.54	0.06	NA	150
DESTN BIT 15	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 14	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 13	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 12	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 11	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 10	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 9	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 8	NA	0.4	2.4	-0.22	0.04	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DESTN BIT 7	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 6	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 5	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 4	NA	0.4	2.4	-0.22	0.04	NA	100
DESTN BIT 3	NA	0.4	2.4	-0.22	0.04	NA	100
DESTN BIT 2	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 1	NA	0.4	2.4	-0.22	0.04	NA	75
DESTN BIT 0	NA	0.4	2.4	-0.22	0.04	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ALU CR IN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
ALU CR P	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
ADR BIT 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
ADR BIT 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	175
ADR BIT 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	175
ADR BIT 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	175
ADR BIT 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
ADR BIT 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
ADR BIT 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
ADR BIT 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
ADR BIT 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
SCE BIT 15	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	125
SCE BIT 14	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	125
SCE BIT 13	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	125

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SCE BIT 12	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	125
SCE BIT 11	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	125
SCE BIT 10	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	125
SCE BIT 9	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 8	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 7	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 6	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 5	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 4	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 3	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	175
SCE BIT 2	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	175
SCE BIT 1	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	150
SCE BIT 0	2.4	0.5	-0.16	1.6	0.08	0.08	-30.0	175
EN TAB UP	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BIT 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
DESTN BIT 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
DESTN BIT 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BIT 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
DESTN BIT 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	PH S/D CLK to SCE BIT 15	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 14	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 13	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 12	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 11	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 10	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 09	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 08	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 07	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 06	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 05	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 04	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 03	-	-	-	107	107	107	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	PH S/D CLK to SCE BIT 02	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 01	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 00	-	-	-	107	107	107	nS	12
2	tpd	PH S/D CLK to DESTN BIT 15	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 14	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 13	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 12	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 11	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 10	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 09	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 08	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 07	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 06	-	-	-	95	95	95	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	PH S/D CLK to DESTN BIT 05	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 04	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 03	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 02	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 01	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 00	-	-	-	95	95	95	nS	12
3	tpd	PH S/D CLK to DESTN BIT 15	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 14	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 13	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 12	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 11	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 10	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 09	-	-	-	146	146	146	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	PH S/D CLK to DESTN BIT 08	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 07	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 06	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 05	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 04	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 03	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 02	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 01	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 00	-	-	-	146	146	146	nS	12
4	tpd	PH S/D CLK to DESTN BIT 15	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 14	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 13	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 12	-	-	-	182	182	182	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	PH S/D CLK to DESTN BIT 11	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 10	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 09	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 08	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 07	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 06	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 05	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 04	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 03	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 02	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 01	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 00	-	-	-	182	182	182	nS	12
5	tsu	DESTN BIT 15 to PH S/D CLK	29	29	29	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	DESTN BIT 14 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 13 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 12 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 11 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 10 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 09 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 08 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 07 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 06 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 05 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 04 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 03 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 02 to PH S/D CLK	29	29	29	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	DESTN BIT 01 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 00 to PH S/D CLK	29	29	29	-	-	-	nS	12
6	tsu	DESTN BIT 15 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 14 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 13 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 12 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 11 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 10 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 09 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 08 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 07 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 06 to PH S/D CLK	46	46	46	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tsu	DESTN BIT 05 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 04 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 03 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 02 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 01 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 00 to PH S/D CLK	46	46	46	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 15	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 14	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 13	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 12	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 11	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 10	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 09	13	13	13	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	th	PH S/D CLK to DESTN BIT 08	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 07	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 06	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 05	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 04	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 03	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 02	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 01	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 00	13	13	13	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 15	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 14	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 13	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 12	2	2	2	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	PH S/D CLK to DESTN BIT 11	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 10	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 09	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 08	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 07	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 06	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 05	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 04	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 03	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 02	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 01	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 00	2	2	2	-	-	-	nS	12

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6.4 Module function. The multiply/math pac (DWAE) module contains the support logic that allows the Central Processor (CP) to execute an additional set of arithmetic instructions. The math pac instructions are identified in the macroinstruction repertoire. The math pac instructions perform the following functions:

(a) Fixed and floating point square root.

(b) Trigonometric and hyperbolic vector and rotate.

(c) Fixed point to floating point conversion, floating point to fixed point conversion, floating point normalize, floating point compare, and quadruple shifts.

(d) Floating point exponential, natural log, sine, cosine, tangent, arcsine, arccosine, and arctangent.

(e) Floating point add, subtract, multiply, and divide.

(f) Double precision multiply and divide.

The multiply/math pac module is standard in the high performance AN/UYK-44 CP. The module also provides high performance CP support logic to decrease execution time of the count ones, set bit, clear bit, compare bit, multiply, and divide instructions. The multiply/math pac module contains a multiplier/accumulator that performs a 16 x 16 bit multiplication and product accumulation, a 2048 32-bit math table Programmable Read Only Memory (PROM) containing constants used for math pac operations, and a floating point characteristic Arithmetic Logic Unit (ALU) to calculate the characteristic for floating point operations.

6.4.1 Multiply circuits. The multiply circuits consist of the multiplier/accumulator, Y holding register, and associated latches that hold the product. The multiplier/accumulator performs 126 x 16 bit multiplication and product accumulation. The result is a 35-bit product/sum divided into Least Significant Product (LSP) (bits 15-0), Most Significant Product (MSP) (bits 31-16), and Extended Product (XTP) (bits 34-32). The multiplier/accumulator contains a 16-bit X input register, a 16-bit Y input register, and a 35-bit product/accumulator register (holds LSP, MSP, and XTP). The X and Y registers hold the operands. The X register is loaded with data from the CP source B/destination bus. Data inputs to the Y register are time-shared with the LSP outputs. The Y register is loaded with data from the Y holding register. Both the X and Y operands can

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be transferred to the multiplier on the same microcycle. The Y operand is sent during source time via the CP source B/destination bus and is captured in the Y holding register. At destination time, the X operand is sent and together with the Y operand (from the Y holding register) is loaded into the multiplier/accumulator. The multiplier/accumulator is controlled by a format 3 microinstruction with Micro Control (MC) = 2. Bits 7-0 are used for multiply operations and are interpreted as follows:

(a) Bits 1 and 0 - Bit 1 enables or disables the accumulator and bit 0 determines if the contents of the product/accumulator is added to or subtracted from the product when the accumulator is enabled.

(b) Bit 2 - When bit 2 equals 1, rounding is enabled. A 1 is added to the most significant bit (bit 15) of LSP in the multiplier to round up the product in MSP and XTP rather than truncate it.

(c) Bit 3 - When bit 3 equals 0, the X and Y inputs are 16-bit unsigned magnitude numbers. When bit 3 equals 1, X and Y inputs are 16-bit two's complement numbers.

(d) Bits 4 and 5 - When bits 5 and 4 equal 11 (base 2), the X and Y registers are loaded and bits 3-0 are interpreted. The X and Y operands are transferred via the source B/destination bus on the same microcycle. The Y operand is transferred during source time and held in the Y holding register. The X operand is transferred during destination time. Both operands are clocked into the multiplier at the end of the microcycle.

(e) Bits 7 and 6 - When bits 7 and 6 equal 01 (base 2) or 10 (base 2), the product/accumulator register is loaded with the result of the multiply/accumulate operation (can occur at earliest on microcycle following loading of X and Y operands). Also when bits 7 and 6 equal 01 (base 2), the left shift by 1 (zero fill) path is enabled for sourcing and product outputs via the source A bus. The left shift mode is enabled until a clock product register operation with bits 7 and 6 equal 10 (base 2) is executed. When bits 7 and 6 equal 11 (base 2), a transfer operation is performed that transfers the XTP (with sign extended) and MSP results to the MSP and LSP product/accumulator register (effectively shifting the result right by 16 bits). The transfer is accomplished via the XTP transfer hold latch and the MSP transfer hold latch.

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(f) Bits 3-0 are only interpreted on the same microcycle that loads both operands into the X and Y registers.

Several microcycles are required to complete a multiply operation. A typical multiply operation would consist of the following sequence:

(a) Load Y and X operands (Y at source, X at destination time) and specify operation in bits 3-0.

(b) Load product/accumulator register with result.

(c) Transfer LSP (microinstruction with Source B (SB) = B) via source A bus.

(d) Transfer MSP (microinstruction with SB = A) via source A bus.

The 16-bit Y holding register holds the Y operand to permit transferring two operands via the source B/destination bus during one microcycle. The register is loaded at the end of the microcycle source time. The register output is enabled by a microinstruction with MC = 2, bits 7 and 6 not equal to 11 (base 2), and bit 4 = 1. The 16-bit XTP and MSP transfer hold latches hold data for a transfer operation. The latches are loaded at the end of source time. XTP bits 2-0 are loaded into XTP transfer hold latch bits 2-0 with bit 2 extended to bits 15-3. Both latches outputs are enabled by a microinstruction with MC = 2 and bits 7 and 6 = 11 (base 2). The data is loaded into the product/accumulator register at the end of the microcycle. Four latches are used to transfer the product via the source A bus (contents of one 16-bit latch per transfer). All latches are loaded at source time. Two latches hold the MSP and LSP. The other two latches hold the MSP and LSP left shifted one bit position with zero fill. The microinstruction SA field selects MSP (SA = A) or LSP (SA = B) and the left shift mode determines if the left shifted latch is selected.

6.4.2 Math table. The math table section consists of math table PROM containing 2048 32-bit words and associated addressing circuits. The PROM contains the following constants:

(a) Series expansion coefficients for integer fixed point transcendental (referred to as CORDIC) and floating point transcendental instructions (1539 32-bit words).

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(b) Division inverse table for double divide and floating point divide instructions (256 16-bit words).

(c) Bit masks for set/clear/test bit instructions (32 16-bit words).

(d) Look-up table for count ones' instruction (256 16-bit words).

Math table operation is controlled by a format 3 micro-instruction with MC = 2. Micro Control Field (MCU) bits 11-8 determine the operation. The math table addressing circuits consist of a math table address upper register (bits 15-8), address lower register (bits 11-4), address register and counter are loaded from the CP source B/destination bus by a microinstruction with D = 04. The counter is incremented by a format 3 microinstruction with MC = 2 and bit 8 = 1. The 11-bit PROM address is formed by the addressing circuits depending on the operation specified by MCU bits 11-9. For a select division inverse table or enable count ones' data operation, the count ones'/divide address multiplexer selects ones' for address bits 10-8 to address the division inverse table or count ones' look-up table in the 7XX address range. Address bits 7-0 are supplied from either the math table address upper register (upper byte) or address lower register bits 7-4 and the address counter (lower byte). For an enable math table lower data or enable math table upper data operation, the PROM is addressed by the lower 11 bits from the math table address register. Selecting an operation with bit 11 = 1, powers up the associated math table PROM. The PROM remains selected and powered up until a microinstruction with SB = 1E is executed to transfer the PROM output data via the CP source B/destination bus. The select division inverse table operation requires that a math table operation with bit 11 = 1 be performed prior to the division inverse table operation to power-up the PROM. A typical math table operation would consist of the following sequence:

(a) Load or increment address.

(b) Select math table operation.

(c) Transfer math table data (16-bits) via source B/destination bus.

6.4.3 Floating point characteristics circuits. These circuits of registers hold floating point characteristics of two operands, a normalize count register to hold the hexadecimal

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normalize count, a bias adjust circuit, an 8-bit ALU, and an output latch. The floating point characteristic 1 register holds one characteristic for a floating point operation. It is loaded with source B/destination bus bits 14-8 by a microinstruction with D = 14 (shadowed behind microaccumulator 4). Its output is always available to the ALU. The floating point characteristic 2 register holds the other characteristic for a floating point operation. It is loaded with source B/destination bus bits 14-8 by a microinstruction with D = 16 (shadowed behind microaccumulator 6). Its output is enabled by a format 3 microinstruction with MC = 3 and bit 4 = 0. The normalize count register converts the binary normalize shift count from the shift matrix into an aligned hexadecimal normalized count. The register is loaded from the source B/destination bus bits 9-2 by a microinstruction with D = 16 (shadowed behind microaccumulator 6). Its output is enabled by a format 3 microinstruction with MC = 3 and bit 4 = 1. The ALU and bias adjust circuit are used to calculate the characteristic for floating point operations. The ALU and bias adjust circuit are controlled by a microinstruction with MC = 3 and bits 7-3. When bits 7-5 = 001 (base 2), the ALU subtracts characteristic 2 (bits 4 = 0) or the normalize count (bit 4 = 1) from characteristic 1. When bits 7-5 = 011 (base 2), the ALU adds characteristic 1 to characteristic 2 or the normalize count. The bias adjust circuit removes the bias from characteristic 2 when bits 4 and 3 = 01 (base 2). Bias is removed prior to ALU operation to provide the desired result for the floating point operation. (Characteristic is always biased at start of operation.) The ALU performs the following floating point operations with the result, and characteristic 2 adjusted as indicated:

(a) Floating point multiply: Biased characteristic 1 plus unbiased (with bias removed) characteristic 2 = biased sum.

(b) Floating point divide: Biased characteristic 1 minus unbiased (with bias removed) characteristic 2 = biased difference.

(c) Floating point add or subtract: Biased characteristic 1 minus biased characteristic 2 = unbiased difference (magnitude comparison).

(d) Floating point normalize: Biased characteristic 1 minus unbiased normalize count = biased difference. (Characteristic 1 is unnormalized result characteristic).

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The floating point characteristic ALU output is loaded into characteristic output latch bits 15-8 with zero fill. The latch output is placed on the source B/destination bus by a microinstruction with SB = 1E. Bit 15 is used as an overflow/ underflow detect flag for floating point multiply, divide, and normalize operations. If bit 15 = 1 and bit 14 = 0, an overflow occurred. If bit 15 = 1 and bit 14 = 1, an underflow occurred. For floating point add subtract operations, bit 15 is used to compare magnitude. If bit 15 = 0, characteristic 1 is equal to or greater than characteristic 2. If bit 15 = 1, characteristic 1 is less than characteristic 2.

6.4.4 Math pac instructions. The math pac instructions add the following functions to the MRP: integer square root, quadruple (64-bit) shifts, floating point arithmetic, floating point transcendentals, and fixed point transcendentals.

6.4.4.1 Integer square root. The integer square root function is implemented as an Op code 10 instruction with an m value of zero. The square root is taken of a double-length number Ra, Ra+1. The number must be positive and cannot be greater than 3FFFFFFF. A negative number or any larger positive number will set the overflow designator (status register 1, bit 10) and the original operands are returned unchanged. At completion, the square root will be in Ra+1 and the remainder will be in Ra as an unsigned magnitude quantity (all 16 bits may be significant). The remainder is defined as the difference between the original double-length number and the square of the root.

6.4.4.2 Double precision multiply and divide. The double precision multiply function (Op Codes B8, B9, BB) multiplies the double-length number specified by the m-field (Rm, Rm+1, or Y, Y+1) by the double-length number Ra, Ra+1. At completion, the result (product) is in Ra, R+1, Ra+2, and Ra+3. The double precision divide function (Op Codes BC, BD, BF) divides the number specified by the m-field (Rm, Rm+1, or Y, Y+1). At completion, the double-length quotient is in Ra+2, Ra+3, and the double-length remainder is in Ra, Ra+1. The remainder has the same sign as the dividend.

6.4.4.3 Quadruple shifts. The algebraic left and right quadruple shift functions (Op Code 7C with m values or E for left shift and F for right shift) shift the contents of Ra, Ra+1, Ra+2, and Ra+3 by the 6 least significant bits of Y. For left shifts, zero fill is performed and for right shifts, sign fill is performed.

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6.4.4.4 Floating point arithmetic. The basic floating point instructions consist of addition (Op Codes A4, A5, A7), subtraction (Op Codes A0, A1, A3), multiplication (Op Codes A8, A9, AB), division (Op Codes AC, AD, AF), comparison (Op Code 7C with m = 8), normalization (Op Code 7C with m = 9), and conversion from floating point to fixed point (Op Code 7C with m = A). Floating point operations use hexadecimal normalized double-length words consisting of a fractional mantissa and a characteristic (exponent). The mantissa is a signed magnitude fraction represented in six 4-bit hexadecimal digits with a radix point to the left of the most significant digit. Bit 15 of the first work is the sign; a zero indicates a positive fraction and a one indicates a negative fraction. The characteristic is a 7-bit quantity representing the signed exponent plus 64 (biased). The magnitude range of normalized floating point numbers is from .100000 (base 16) x (16 to the -64) to .FFFFFF (base 16) x (16 to the +63) which is approximately 5.4 x (10 to the -79) to 7.2 x (10 to the 75) decimal. A normalized floating point zero is represented by a zero sign, zero characteristic and zero fraction. A non-zero floating point number is normalized when the most significant hexadecimal digit is non-zero. Results (excluding residue) of all floating point operations are always normalized regardless of the input operands. The remainder (often called the residue) resulting from floating point add, subtract, and multiply operations, consists of those digits from the least significant portion of the result that cannot be contained within six hexadecimal digits. The remainder resulting from a floating point divide operation is the undivided portion of the dividend which is less than the divisor and remains when the division process is completed. It has the same sign as the dividend and will be performed on floating-point operations when specified by bit 6 of status register 1. When residue is specified, the MRP forms the remainder into a floating-point number in Ra+2 and Ra+3. When bit 6 is clear, no residue is generated and the contents of Ra+2 and Ra+3 remain unchanged. A floating point causes a characteristic underflow condition whenever a number smaller than .100000 x (16 to the -64) is generated as the result. A floating point operation causes a characteristic overflow condition whenever a number larger than .FFFFF (base 16) x (16 to the +63) is generated as the result. No underflow condition occurs in forming a residue. If a residue number smaller than .100000 (base 16) x (16 to the -64) is generated, the

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residue value is all zeros. When a floating point operation causes a characteristic underflow or overflow, the MRP performs the following:

(a) Inhibits storing the erroneous result in the general registers.

(b) Modifies the Condition Code of Status Register 1. Set bits 10 and 11 if underflow. Set bit 10 and clear bit 11 if overflow.

(c) Modify Status Register 2. Store Instruction Register bits 11-4 into Status Register 2 bits 7-0 if bit 7 in Status Register 1 is zero.

(d) Test bit 7 and bit 2 of Status Register 1. If bit 7 is a 1, disable the interrupt independent of the status of bit 2; if bit 7 is a 0 and bit 2 is a 1, set the floating-point interrupt and process; if bit 7 is a 0 and bit 2 is a 0, disable the interrupt.

The original contents of the general registers and memory addresses Y and Y+1 remain unchanged.

6.4.4.5 Floating point transcendental functions. The floating point transcendental functions include square root, sine, cosine, tangent, arcsine, arccosine, arctangent, exponential, and natural log. All functions except square root are implemented by a 7D instruction. Square root is implemented by a 10 instruction with m = 5. The instruction work is single-length. The a-designator specifies the first of two sequential general registers that contain the X value expressed in radian units. Transcendental functions are solved using a modified Taylor series expansion expressed as follows:

$$Y = F(Xo) + \frac{F'(Xo)(dX)}{1!} + \frac{F''(Xo)((dX)^2)}{2!} + \frac{F'''(Xo)((dX)^3)}{3!}$$

where: Xo is expansion point
dX is distance from expansion point
F'(X), F"(X), F'"(X) are derivatives of F(X)
^ is the symbol for exponentiation

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Modified series can be written as:

$$Y = (((((dX\ .01)+02).dX)+C3).dX)+C4$$

where: $C1 = \frac{F'''(Xo)}{3!}$

$C2 = \frac{F''(Xo)}{2!}$

$C3 = \frac{F'(Xo)}{1!}$

$C4 = F(Xo)$

Coefficients C1, C2, C3, and C4 are constants and are stored in the math table PROM. To solve the function, X must be in the domain of series expansion. The series expansion domain is divided into 32 intervals (expansion points) with four coefficients stored in the math table PROM for each interval. If X is not within the domain, a transformation is used to place X in the domain. In either case, a new X is formed with sign bit and characteristic removed. Bits 31 through 27 contain the interval number that is used to address the four coefficients stored in the math table PROM and bits 26 through 0 contain the delta X that is used in the Taylor series expansion. Values outside the function domain will set the underflow/overflow bits (status register 1 bits 11 and 10).

6.4.4.6 Fixed point transcendental (CORDIC) functions. The fixed point transcendental (commonly referred to as CORDIC) provide the capability for efficient handling of trigonometric and hyperbolic computations. CORDIC functions are initiated by a 7C instruction. The instruction provides trigonometric vector and rotate, and hyperbolic vector and rotate functions, with or without scaling. The CORDIC instruction word is single-length. The a-designator specifies that first of three sequential general registers. The register specified (Ra) contains the Y-coordinate value and is referred to as Y. Ra+1 contains the x-coordinate value and is referred to as X. Ra+2 contains the angle value and is referred to as W. M-designator values 0-3 select trigonometric functions and m-designator values 4-7 hyperbolic functions.

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Coordinate values must be properly scaled to prevent erroneous results caused by overflow. The overflow bit (status register 1 bit 0) is set when overflow occurs on a CORDIC instruction. Coordinate data is represented by 16-bit values with bit 15 the sign bit. Negative quantities are in two's complement. Angles are represented in Binary Angular Measurement (BAMS). For angular data, bit 15 represents 180 degrees, and each less significant bit represents an angle one-half the value of the previous bit position. The value of the least significant bit is 180 degrees/2 to the 15th power or 0.00549 degrees. The binary point for the angle is between bits 15 and 14. The scale sequence corrects for an error inherent in the CORDIC process by multiplying the X and Y values by a constant. The constant for trigonometric scaling is 0.46672 (base 8) and for hyerbolic scaling is 1.15217 (base 8). Scaling is performed when m equals 2, 3, 6, or 7. For trigonometric functions, scaling occurs before the compute sequence (prescaling), and for hyperbolic functions scaling occurs after the compute sequence (post-scaling). The compute sequence produces the desired trigonometric or hyperbolic solution of shifts, arithmetic calculations, and a modified Taylor series expansion expressed as follows:

$$Y = (((C2.\ X)+C3)(\ X))+C4$$

Paragraph 6.4.4.5 describes the modified Taylor series expansion. For CORDIC instructions, X is 16 bits, with bits 15 through 11 containing the interval number and bits 10 through delta X.

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6.4.5 Description of I/O pins.

Signal name	Pin number	I/O	Description
ALU CR IN	2	I/O	FP CHAR ALU CR IN
ALU GR P	3	I/O	FP CHAR ALU GR P
ADR BIT 8	4	I/O	MATH TABLE ADRS BIT 08
IR BIT 0	5	I	CP MICRO IR BIT 00
IR BIT 1	6	I	CP MICRO IR BIT 01
IR BIT 2	7	I	CP MICRO IR BIT 02
IR BIT 3	11	I	CP MICRO IR BIT 03
ADR BIT 7	12	I/O	MATH TABLE ADRS BIT 07
ADR BIT 6	13	I/O	MATH TABLE ADRS BIT 06
ADR BIT 5	14	I/O	MATH TABLE ADRS BIT 05
IR BIT 4	21	I	CP MICRO IR BIT 4
IR BIT 5	22	I	CP MICRO IR BIT 5
IR BIT 6	23	I	CP MICRO IR BIT 6
IR BIT 7	24	I	CP MICRO IR BIT 7
IR BIT 8	26	I	CP MICRO IR BIT 8
PH S/D CLK	27	I	PH S/D CLK
SCE A EN	28	I	PH SCE A ENBL
IR BIT 40	29	I	CP MICRO IR BIT 40
IR BIT 23	30	I	CP MICRO IR BIT 23
IR BIT 22	32	I	CP MICRO IR BIT 22

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IR BIT 21	33	I	CP MICRO IR BIT 21
SCE BIT 15	34	O	CP SCE A BUS BIT 15
SCE BIT 14	35	O	CP SCE A BUS BIT 14
SCE BIT 13	36	O	CP SCE A BUS BIT 13
SCE BIT 12	37	O	CP SCE A BUS BIT 12
SCE BIT 11	38	O	CP SCE A BUS BIT 11
SCE BIT 10	39	O	CP SCE A BUS BIT 10
SCE BIT 9	41	O	CP SCE A BUS BIT 9
SCE BIT 8	42	O	CP SCE A BUS BIT 8
SCE BIT 7	43	O	CP SCE A BUS BIT 7
SCE BIT 6	44	O	CP SCE A BUS BIT 6
SCE BIT 5	45	O	CP SCE A BUS BIT 5
SCE BIT 4	46	O	CP SCE A BUS BIT 4
SCE BIT 3	47	O	CP SCE A BUS BIT 3
SCE BIT 2	48	O	CP SCE A BUS BIT 2
SCE BIT 1	49	O	CP SCE A BUS BIT 1
SCE BIT 0	50	O	CP SCE A BUS BIT 0
ADR BIT 4	52	I/O	MATH TABLE ADR BIT 4
ADR BIT 3	53	I/O	MATH TABLE ADR BIT 3
ADR BIT 2	54	I/O	MATH TABLE ADR BIT 2

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ADR BIT 1	55	I/O	MATH TABLE ADRS BIT 1
ADR BIT 0	56	I/O	MATH TABLE ADRS BIT 0
EN ADR UP	57	I	ENABLE TABLE ADRS UPPER (IN)
EN TAB UP	58	O	ENBL TABLE ADRS UPPER (OUT)
DESTN BIT 15	59	I/O	CP SCE B/DESTN BUS BIT 15
DESTN BIT 14	61	I/O	CP SCE B/DESTN BUS BIT 14
DESTN BIT 13	62	I/O	CP SCE B/DESTN BUS BIT 13
DESTN BIT 12	63	I/O	CP SCE B/DESTN BUS BIT 12
DESTN BIT 11	64	I/O	CP SCE B/DESTN BUS BIT 11
DESTN BIT 10	65	I/O	CP SCE B/DESTN BUS BIT 10
DESTN BIT 9	66	I/O	CP SCE B/DESTN BUS BIT 9
DESTN BIT 8	67	I/O	CP SCE B/DESTN BUS BIT 8
DESTN BIT 7	68	I/O	CP SCE B/DESTN BUS BIT 7
DESTN BIT 6	69	I/O	CP SCE B/DESTN BUS BIT 6
DESTN BIT 5	70	I/O	CP SCE B/DESTN BUS BIT 5
DESTN BIT 4	71	I/O	CP SCE B/DESTN BUS BIT 4
DESTN BIT 3	72	I/O	CP SCE B/DESTN BUS BIT 3
DESTN BIT 2	73	I/O	CP SCE B/DESTN BUS BIT 2
DESTN BIT 1	74	I/O	CP SCE B/DESTN BUS BIT 1
DESTN BIT 0	76	I/O	CP SCE B/DESTN BUS BIT 0

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EN M ADR BIT	77	I	ENABLE MATH TABLE ADRS BITS
MACLR SYNC	78	I	CP MACLR SYNC
IR BIT 11	79	I	CP MICRO IR BIT 11
IR BIT 10	80	I	CP MICRO IR BIT 10
IR BIT 9	82	I	CP MICRO IR BIT 9
IR BIT 20	83	I	CP MICRO IR BIT 20
SCE B EN	84	I	PH SCE B ENBL
IR BIT 16	85	I	CP MICRO IR BIT 16
IR BIT 19	86	I	CP MICRO IR BIT 19
IR BIT 18	87	I	CP MICRO IR BIT 18
IR BIT 17	88	I	CP MICRO IR BIT 17
CONT FRMT	89	I	MICRO CONT FORMAT
IR BIT 15	91	I	CP MICRO IR BIT 15
IR BIT 14	92	I	CP MICRO IR BIT 14
IR BIT 13	93	I	CP MICRO IR BIT 13
IR BIT 12	94	I	CP MICRO IR BIT 12
IR BIT 27	96	I	CP MICRO IR BIT 27
IR BIT 24	97	I	CP MICRO IR BIT 24
IR BIT 28	98	I	CP MICRO IR BIT 28
IR BIT 26	99	I	CP MICRO IR BIT 26

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IR BIT 25	100	I	CP MICRO IR BIT 25

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N717)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,8,9,15, 16,17,18, 19,20	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	8,9,15, 16,17,18, 19,20	C				
A3	Isolation, unused to unused pins	8	9,15,16, 17,18, 19,20				
A4	Isolation, unused to unused pins	9	15,16, 17,18, 19,20				
A5	Isolation, unused to unused pins	15	16,17, 18,19,20				
A6	Isolation, unused to unused pins	16	17,18, 19,20				
A7	Isolation, unused to unused pins	17	18,19,20				
A8	Isolation, unused to unused pins	18	19,20				
A9	Isolation, unused to unused pins	19	20	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A9
Isolation measurement	NA	± 500 KOhm	A1-A9

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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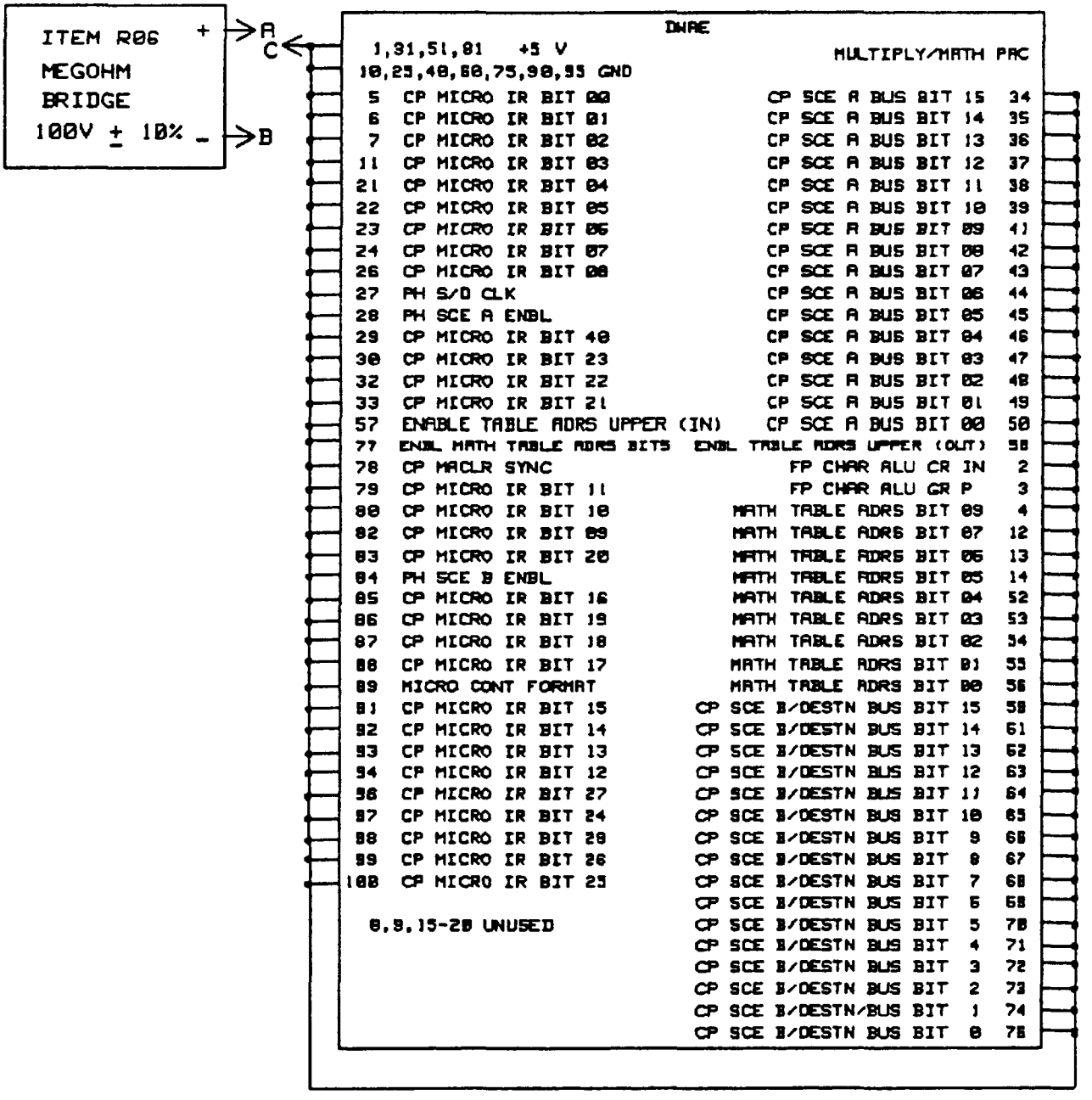


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, GND	10	60				
B6	Continuity, GND	10	95				
B7	Continuity, Vcc	1	31				
B8	Continuity, Vcc	51	81				
B9	Continuity, Vcc	1	51	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B9

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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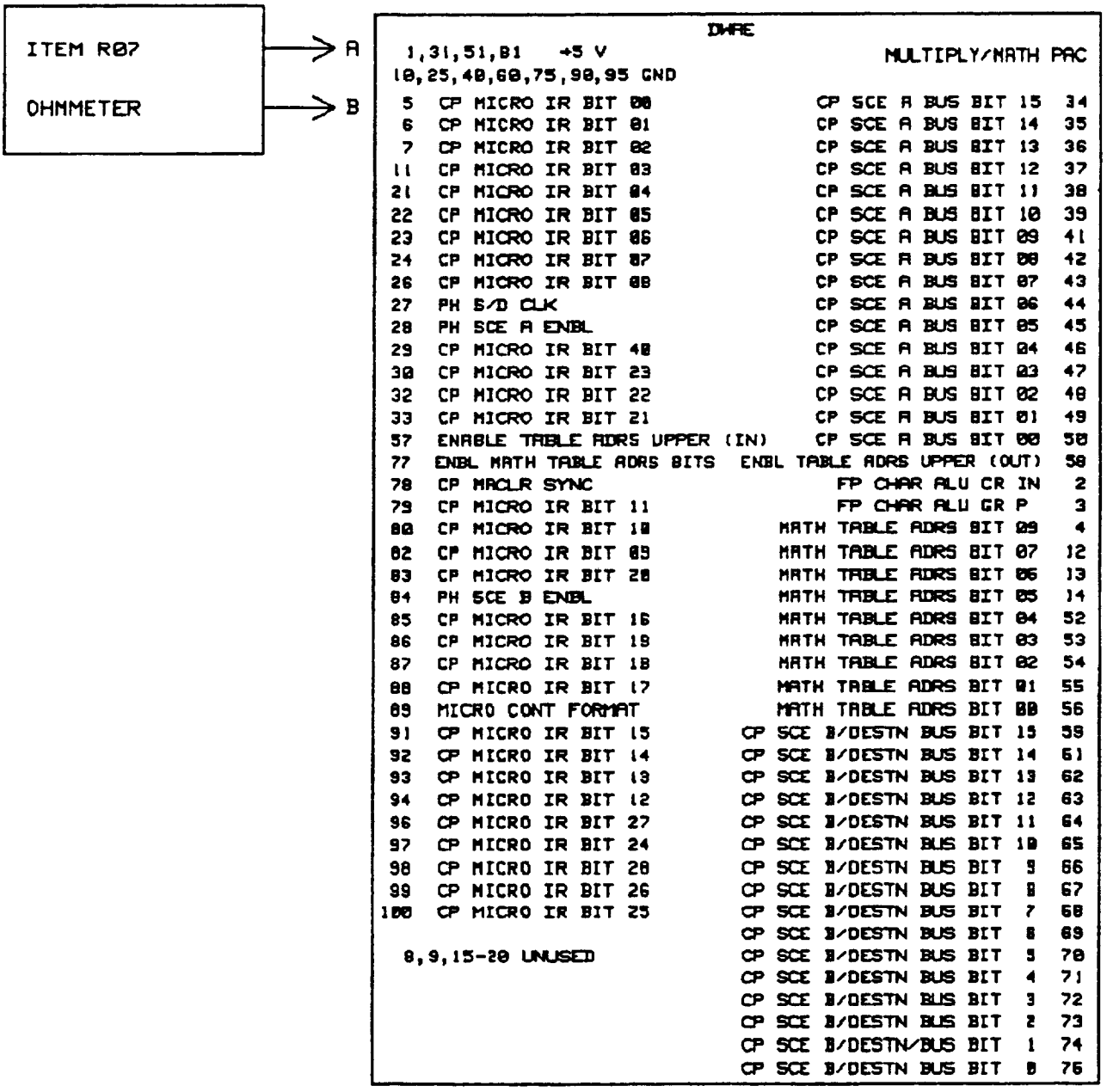


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Connect		25 ± 5°C Initial limits	Notes
		Positive lead to	Negative lead to		
C	Vcc stress	1,31, 51,81	10,25, 40,60, 75,90, 95	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	4.5 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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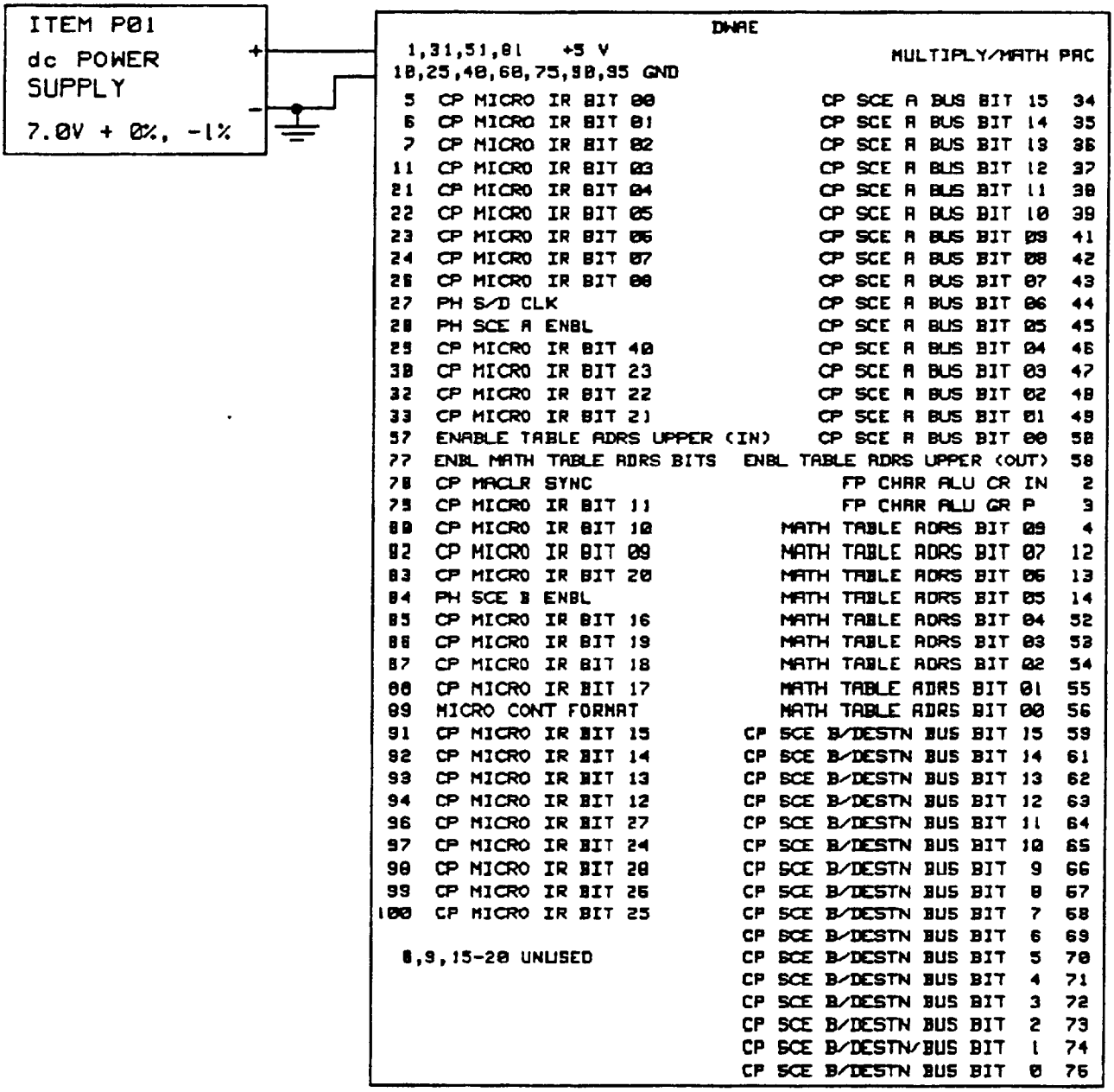


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	IR BIT 0	5	5.5	0.0	1.0	mA	1-3
D2	IR BIT 1	6					
D3	IR BIT 2	7					
D4	IR BIT 3	11	5.5	0.0	1.0	mA	1-3
D5	IR BIT 4	21	5.5	0.0	0.3	mA	1-3
D6	IR BIT 5	22	5.5	0.0	0.3	mA	1-3
D7	IR BIT 6	23	5.5	0.0	0.4	mA	1-3
D8	IR BIT 7	24	5.5	0.0	0.3	mA	1-3
D9	IR BIT 8	26	5.5	0.0	1.0	mA	1-3
D10	PH S/D CLK	27	5.5	0.0	8.4	mA	1-3
D11	SCE A EN	28	5.5	0.0	0.1	mA	1-3
D12	IR BIT 40	29					
D13	IR BIT 23	30					
D14	IR BIT 22	32					
D15	IR BIT 21	33					
D16	EN ADR UP	57					
D17	EN M ADR BIT	77	5.5	0.0	0.1	mA	1-3
D18	MACLR SYNC	78	5.5	0.0	0.4	mA	1-3
D19	IR BIT 11	79	5.5	0.0	0.2	mA	1-3
D20	IR BIT 10	80	5.5	0.0	0.2	mA	1-3
D21	IR BIT 9	82	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D22	IR BIT 20	83	5.5	0.0	0.1	mA	1-3
D23	SCE B EN	84					
D24	IR BIT 16	85					
D25	IR BIT 19	86					
D26	IR BIT 18	87					
D27	IR BIT 17	88					
D28	CONT FRMT	89					
D29	IR BIT 15	91					
D30	IR BIT 14	92					
D31	IR BIT 13	93					
D32	IR BIT 12	94					
D33	IR BIT 27	96					
D34	IR BIT 24	97					
D35	IR BIT 28	98					
D36	IR BIT 26	99					
D37	IR BIT 25	100	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D37
Power supply current limit	3.75 A	NA	D1-D37
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D37
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D37
low, (VL)	0.0 V	± 100 mV	D1-D37
Input current measurement	NA	± 50 uA	D1-D4
	NA	± 15 uA	D5,D6
	NA	± 20 uA	D7
	NA	± 15 uA	D8
	NA	± 50 uA	D9
	NA	± 420 uA	D10
	NA	± 5 uA	D11-D17
	NA	± 20 uA	D18
	NA	± 10 uA	D19,D20
	NA	± 5 uA	D21-D37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	ALU CR IN	2	0.0	-150.0	-60.0	mA	1-4
E2	ALU GR P	3					
E3	ADR BIT 8	4	0.0	-150.0	-60.0	mA	1-4
E4	ADR BIT 7	12	0.0	-260.0	-30.0	mA	1-4
E5	ADR BIT 7	12					
E6	ADR BIT 6	13					
E7	ADR BIT 6	13					
E8	ADR BIT 5	14					
E9	ADR BIT 5	14	0.0	-260.0	-30.0	mA	1-4
E10	SCE BIT 15	34	0.0	-220.0	-30.0	mA	1-4
E11	SCE BIT 15	34					
E12	SCE BIT 15	34					
E13	SCE BIT 15	34					
E14	SCE BIT 14	35					
E15	SCE BIT 14	35					
E16	SCE BIT 14	35					
E17	SCE BIT 14	35					
E18	SCE BIT 13	36					
E19	SCE BIT 13	36					
E20	SCE BIT 13	36					
E21	SCE BIT 13	36	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E22	SCE BIT 12	37	0.0	-220.0	-30.0	mA	1-4
E23	SCE BIT 12	37					
E24	SCE BIT 12	37					
E25	SCE BIT 12	37					
E26	SCE BIT 11	38					
E27	SCE BIT 11	38					
E28	SCE BIT 11	38					
E29	SCE BIT 11	38					
E30	SCE BIT 10	39					
E31	SCE BIT 10	39					
E32	SCE BIT 10	39					
E33	SCE BIT 10	39					
E34	SCE BIT 9	41					
E35	SCE BIT 9	41					
E36	SCE BIT 9	41					
E37	SCE BIT 9	41					
E38	SCE BIT 8	42					
E39	SCE BIT 8	42					
E40	SCE BIT 8	42					
E41	SCE BIT 8	42					
E42	SCE BIT 7	43	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	SCE BIT 7	43	0.0	-220.0	-30.0	mA	1-4
E44	SCE BIT 7	43					
E45	SCE BIT 7	43					
E46	SCE BIT 6	44					
E47	SCE BIT 6	44					
E48	SCE BIT 6	44					
E49	SCE BIT 6	44					
E50	SCE BIT 5	45					
E51	SCE BIT 5	45					
E52	SCE BIT 5	45					
E53	SCE BIT 5	45					
E54	SCE BIT 4	46					
E55	SCE BIT 4	46					
E56	SCE BIT 4	46					
E57	SCE BIT 4	46					
E58	SCE BIT 3	47					
E59	SCE BIT 3	47					
E60	SCE BIT 3	47					
E61	SCE BIT 3	47					
E62	SCE BIT 2	48					
E63	SCE BIT 2	48	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E64	SCE BIT 2	48	0.0	-220.0	-30.0	mA	1-4
E65	SCE BIT 2	48					
E66	SCE BIT 1	49					
E67	SCE BIT 1	49					
E68	SCE BIT 1	49					
E69	SCE BIT 1	49					
E70	SCE BIT 0	50					
E71	SCE BIT 0	50					
E72	SCE BIT 0	50	0.0	-220.0	-30.0	mA	1-4
E73	ADR BIT 4	52	0.0	-260.0	-30.0	mA	1-4
E74	ADR BIT 4	52					
E75	ADR BIT 3	53	0.0	-260.0	-30.0	mA	1-4
E76	ADR BIT 3	53	0.0	-220.0	-30.0	mA	1-4
E77	ADR BIT 2	54	0.0	-260.0	-30.0	mA	1-4
E78	ADR BIT 2	54	0.0	-220.0	-30.0	mA	1-4
E79	ADR BIT 1	55	0.0	-260.0	-30.0	mA	1-4
E80	ADR BIT 1	55	0.0	-220.0	-30.0	mA	1-4
E81	ADR BIT 0	56	0.0	-260.0	-30.0	mA	1-4
E82	ADR BIT 0	56	0.0	-220.0	-30.0	mA	1-4
E83	EN TAB UP	58	0.0	-150.0	-60.0	mA	1-4
E84	DESTN BIT 15	59	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E85	DESTN BIT 14	61	0.0	-220.0	-30.0	mA	1-4
E86	DESTN BIT 13	62					
E87	DESTN BIT 12	63					
E88	DESTN BIT 11	64					
E89	DESTN BIT 10	65					
E90	DESTN BIT 9	66					
E91	DESTN BIT 8	67					
E92	DESTN BIT 7	68					
E93	DESTN BIT 6	69					
E94	DESTN BIT 5	70					
E95	DESTN BIT 4	71					
E96	DESTN BIT 3	72					
E97	DESTN BIT 2	73					
E98	DESTN BIT 1	74					
E99	DESTN BIT 0	76	0.0	-220.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E99
Power supply current limit	3.75 A	NA	E1-E99
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E99
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E99 E1-E99
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 4.50 mA ± 11.50 mA ± 9.50 mA ± 11.50 mA ± 9.50 mA ± 11.50 mA ± 9.50 mA ± 11.50 mA ± 9.50 mA ± 11.50 mA ± 9.50 mA ± 4.50 mA ± 9.50 mA	E1-E3 E4-E9 E10-E72 E73-E75 E76 E77 E78 E79 E80 E81 E82 E83 E84-E99

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	ALU CR IN	2	2.4	-0.55	0.07	mA	1-3
F2	ALU GR P	3					
F3	ADR BIT 8	4	2.4	-0.55	0.07	mA	1-3
F4	IR BIT 0	5	2.4	-0.4	0.08	mA	1-3
F5	IR BIT 1	6					
F6	IR BIT 2	7	2.4	-0.4	0.08	mA	1-3
F7	IR BIT 3	11	2.4	-1.0	0.10	mA	1-3
F8	ADR BIT 7	12	2.4	-0.54	0.06	mA	1-3
F9	ADR BIT 6	13					
F10	ADR BIT 5	14	2.4	-0.54	0.06	mA	1-3
F11	IR BIT 4	21	2.4	-1.8	0.06	mA	1-3
F12	IR BIT 5	22	2.4	-4.6	0.12	mA	1-3
F13	IR BIT 6	23	2.4	-5.2	0.14	mA	1-3
F14	IR BIT 7	24	2.4	-4.6	0.12	mA	1-3
F15	IR BIT 8	26	2.4	-2.0	0.05	mA	1-3
F16	PH S/D CLK	27	2.4	-17.6	0.48	mA	1-3
F17	SCE A EN	28	2.4	-0.6	0.02	mA	1-3
F18	IR BIT 40	29					
F19	IR BIT 23	30					
F20	IR BIT 22	32	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	IR BIT 21	33	2.4	-0.6	0.02	mA	1-3
F22	ADR BIT 4	52	2.4	-0.54	0.06	mA	1-3
F23	ADR BIT 3	53					
F24	ADR BIT 2	54					
F25	ADR BIT 1	55					
F26	ADR BIT 0	56	2.4	-0.54	0.06	mA	1-3
F27	EN ADR UP	57	2.4	-0.2	0.02	mA	1-3
F28	DESTN BIT 15	59	2.4	-0.22	0.04	mA	1-3
F29	DESTN BIT 14	61					
F30	DESTN BIT 13	62					
F31	DESTN BIT 12	63					
F32	DESTN BIT 11	64					
F33	DESTN BIT 10	65					
F34	DESTN BIT 9	66					
F35	DESTN BIT 8	67					
F36	DESTN BIT 7	68					
F37	DESTN BIT 6	69					
F38	DESTN BIT 5	70					
F39	DESTN BIT 4	71					
F40	DESTN BIT 3	72					
F41	DESTN BIT 2	73	2.4	-0.22	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F42	DESTN BIT 1	74	2.4	-0.22	0.04	mA	1-3
F43	DESTN BIT 0	76	2.4	-0.22	0.04	mA	1-3
F44	EN M ADR BIT	77	2.4	-0.6	0.02	mA	1-3
F45	MACLR SYNC	78	2.4	-4.85	0.08	mA	1-3
F46	IR BIT 11	79	2.4	-1.2	0.04	mA	1-3
F47	IR BIT 10	80	2.4	-1.2	0.04	mA	1-3
F48	IR BIT 9	82	2.4	-0.6	0.02	mA	1-3
F49	IR BIT 20	83					
F50	SCE B EN	84					
F51	IR BIT 16	85					
F52	IR BIT 19	86					
F53	IR BIT 18	87					
F54	IR BIT 17	88					
F55	CONT FRMT	89					
F56	IR BIT 15	91					
F57	IR BIT 14	92					
F58	IR BIT 13	93					
F59	IR BIT 12	94					
F60	IR BIT 27	96					
F61	IR BIT 24	97	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F62	IR BIT 28	98	2.4	-0.6	0.02	mA	1-3
F63	IR BIT 26	99					
F64	IR BIT 25	100	2.4	-0.6	0.02	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F64
Power supply current limit	3.75 A	NA	F1-F64
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F64
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F64
Pattern voltage low, (VL)	0.0 V	± 100 mV	F1-F64
Input current measurement	NA	± 31 uA	F1-F3
	NA	± 24 uA	F4-F6
	NA	± 55 uA	F7
	NA	± 30 uA	F8-F10
	NA	± 93 uA	F11
	NA	± 236 uA	F12
	NA	± 267 uA	F13
	NA	± 236 uA	F14

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 102 uA	F15
	NA	± 900 uA	F16
	NA	± 31 uA	F17-F21
	NA	± 30 uA	F22-F26
	NA	± 11 uA	F27
	NA	± 13 uA	F28-F43
	NA	± 31 uA	F44
	NA	± 246 uA	F45
	NA	± 62 uA	F46,F47
	NA	± 31 uA	F48-F64

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	ALU CR IN	2	0.4	-0.55	0.07	mA	1-3
G2	ALU GR P	3					
G3	ADR BIT 8	4	0.4	-0.55	0.07	mA	1-3
G4	IR BIT 0	5	0.4	-0.4	0.08	mA	1-3
G5	IR BIT 1	6					
G6	IR BIT 2	7	0.4	-0.4	0.08	mA	1-3
G7	IR BIT 3	11	0.4	-1.0	0.10	mA	1-3
G8	ADR BIT 7	12	0.4	-0.54	0.06	mA	1-3
G9	ADR BIT 6	13					
G10	ADR BIT 5	14	0.4	-0.54	0.06	mA	1-3
G11	IR BIT 4	21	0.4	-1.8	0.06	mA	1-3
G12	IR BIT 5	22	0.4	-4.6	0.12	mA	1-3
G13	IR BIT 6	23	0.4	-5.2	0.14	mA	1-3
G14	IR BIT 7	24	0.4	-4.6	0.12	mA	1-3
G15	IR BIT 8	26	0.4	-2.0	0.05	mA	1-3
G16	PH S/D CLK	27	0.4	-17.6	0.48	mA	1-3
G17	SCE A EN	28	0.4	-0.6	0.02	mA	1-3
G18	IR BIT 40	29					
G19	IR BIT 23	30					
G20	IR BIT 22	32					
G21	IR BIT 21	33	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G22	ADR BIT 4	52	0.4	-0.54	0.06	mA	1-3
G23	ADR BIT 3	53					
G24	ADR BIT 2	54					
G25	ADR BIT 1	55					
G26	ADR BIT 0	56	0.4	-0.54	0.06	mA	1-3
G27	EN ADR UP	57	0.4	-0.2	0.02	mA	1-3
G28	DESTN BIT 15	59	0.4	-0.22	0.04	mA	1-3
G29	DESTN BIT 14	61					
G30	DESTN BIT 13	62					
G31	DESTN BIT 12	63					
G32	DESTN BIT 11	64					
G33	DESTN BIT 10	65					
G34	DESTN BIT 9	66					
G35	DESTN BIT 8	67					
G36	DESTN BIT 7	68					
G37	DESTN BIT 6	69					
G38	DESTN BIT 5	70					
G39	DESTN BIT 4	71					
G40	DESTN BIT 3	72					
G41	DESTN BIT 2	73					
G42	DESTN BIT 1	74	0.4	-0.22	0.04	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G43	DESTN BIT 0	76	0.4	-0.22	0.04	mA	1-3
G44	EN M ADR BIT	77	0.4	-0.6	0.02	mA	1-3
G45	MACLR SYNC	78	0.4	-4.85	0.08	mA	1-3
G46	IR BIT 11	79	0.4	-1.2	0.04	mA	1-3
G47	IR BIT 10	80	0.4	-1.2	0.04	mA	1-3
G48	IR BIT 9	82	0.4	-0.6	0.02	mA	1-3
G49	IR BIT 20	83					
G50	SCE B EN	84					
G51	IR BIT 16	85					
G52	IR BIT 19	86					
G53	IR BIT 18	87					
G54	IR BIT 17	88					
G55	CONT FRMT	89					
G56	IR BIT 15	91					
G57	IR BIT 14	92					
G58	IR BIT 13	93					
G59	IR BIT 12	94					
G60	IR BIT 27	96					
G61	IR BIT 24	97	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G62	IR BIT 28	98	0.4	-0.6	0.02	mA	1-3
G63	IR BIT 26	99					
G64	IR BIT 25	100	0.4	-0.6	0.02	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G64
Power supply current limit	3.75 A	NA	G1-G64
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G64
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G64
low, (VL)	0.0 V	± 100 mV	G1-G64
Input current measurement	NA	± 31 uA	G1-G3
	NA	± 24 uA	G4-G6
	NA	± 55 uA	G7
	NA	± 30 uA	G8-G10
	NA	± 93 uA	G11
	NA	± 236 uA	G12
	NA	± 267 uA	G13
	NA	± 236 uA	G14

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NOTES: - Continued.

Condition	Value	Accuracy	Test Condition
Input current measurement	NA	± 102 uA	G15
	NA	± 900 uA	G16
	NA	± 31 uA	G17-G21
	NA	± 30 uA	G22-G26
	NA	± 11 uA	G27
	NA	± 13 uA	G28-G43
	NA	± 31 uA	G44
	NA	± 246 uA	G45
	NA	± 62 uA	G46,G47
	NA	± 31 uA	G48-G64

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	ALU CR IN	2	-0.4	2.4	4.5	V	1-4
H2	ALU GR P	3					
H3	ADR BIT 8	4	-0.4	2.4	4.5	V	1-4
H4	ADR BIT 7	12	-0.16	2.4	4.5	V	1-4
H5	ADR BIT 7	12					
H6	ADR BIT 6	13					
H7	ADR BIT 6	13					
H8	ADR BIT 5	14					
H9	ADR BIT 5	14					
H10	SCE BIT 15	34					
H11	SCE BIT 15	34					
H12	SCE BIT 15	34					
H13	SCE BIT 15	34					
H14	SCE BIT 14	35					
H15	SCE BIT 14	35					
H16	SCE BIT 14	35					
H17	SCE BIT 14	35					
H18	SCE BIT 13	36					
H19	SCE BIT 13	36					
H20	SCE BIT 13	36					
H21	SCE BIT 13	36	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	SCE BIT 12	37	-0.16	2.4	4.5	V	1-4
H23	SCE BIT 12	37					
H24	SCE BIT 12	37					
H25	SCE BIT 12	37					
H26	SCE BIT 11	38					
H27	SCE BIT 11	38					
H28	SCE BIT 11	38					
H29	SCE BIT 11	38					
H30	SCE BIT 10	39					
H31	SCE BIT 10	39					
H32	SCE BIT 10	39					
H33	SCE BIT 10	39					
H34	SCE BIT 9	41					
H35	SCE BIT 9	41					
H36	SCE BIT 9	41					
H37	SCE BIT 9	41					
H38	SCE BIT 8	42					
H39	SCE BIT 8	42					
H40	SCE BIT 8	42					
H41	SCE BIT 8	42					
H42	SCE BIT 7	43	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H43	SCE BIT 7	43	-0.16	2.4	4.5	V	1-4
H44	SCE BIT 7	43					
H45	SCE BIT 7	43					
H46	SCE BIT 6	44					
H47	SCE BIT 6	44					
H48	SCE BIT 6	44					
H49	SCE BIT 6	44					
H50	SCE BIT 5	45					
H51	SCE BIT 5	45					
H52	SCE BIT 5	45					
H53	SCE BIT 5	45					
H54	SCE BIT 4	46					
H55	SCE BIT 4	46					
H56	SCE BIT 4	46					
H57	SCE BIT 4	46					
H58	SCE BIT 3	47					
H59	SCE BIT 3	47					
H60	SCE BIT 3	47					
H61	SCE BIT 3	47					
H62	SCE BIT 2	48					
H63	SCE BIT 2	48	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H64	SCE BIT 2	48	-0.16	2.4	4.5	V	1-4
H65	SCE BIT 2	48					
H66	SCE BIT 1	49					
H67	SCE BIT 1	49					
H68	SCE BIT 1	49					
H69	SCE BIT 1	49					
H70	SCE BIT 0	50					
H71	SCE BIT 0	50					
H72	SCE BIT 0	50					
H73	ADR BIT 4	52					
H74	ADR BIT 4	52					
H75	ADR BIT 3	53					
H76	ADR BIT 3	53					
H77	ADR BIT 2	54					
H78	ADR BIT 2	54					
H79	ADR BIT 1	55					
H80	ADR BIT 1	55					
H81	ADR BIT 0	56					
H82	ADR BIT 0	56	-0.16	2.4	4.5	V	1-4
H83	EN TAB UP	58	-0.4	2.4	4.5	V	1-4
H84	DESTN BIT 15	59	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H85	DESTN BIT 14	61	-0.16	2.4	4.5	V	1-4
H86	DESTN BIT 13	62					
H87	DESTN BIT 12	63					
H88	DESTN BIT 11	64					
H89	DESTN BIT 10	65					
H90	DESTN BIT 9	66					
H91	DESTN BIT 8	67					
H92	DESTN BIT 7	68					
H93	DESTN BIT 6	69					
H94	DESTN BIT 5	70					
H95	DESTN BIT 4	71					
H96	DESTN BIT 3	72					
H97	DESTN BIT 2	73					
H98	DESTN BIT 1	74					
H99	DESTN BIT 0	76	-0.16	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:32Z
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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H99
Power supply current limit	3.75 A	NA	H1-H99
Forcing current, (Ioh)	-0.4mA -0.16mA -0.4mA -0.16mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA	H1-H3 H4-H82 H83 H84-H99
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H99 H1-H99
Output voltage measurement	NA	± 105 mV	H1-H99

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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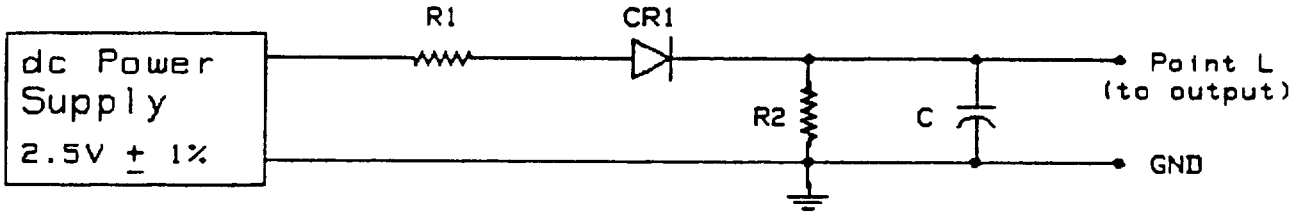
NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,58	6,200	6.0	0.5
12,13,14,34,35, 36,37,38,39,41, 42,43,44,45,46, 47,48,49,50,52, 53,54,55,56,59, 61,62,63,64,65, 66,67,68,69,70, 71,72,73,74,76	16,000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	ALU CR IN	2	6.0	0.0	0.5	V	1-4
J2	ALU GR P	3					
J3	ADR BIT 8	4	6.0	0.0	0.5	V	1-4
J4	ADR BIT 7	12	1.6	0.0	0.5	V	1-4
J5	ADR BIT 7	12					
J6	ADR BIT 6	13					
J7	ADR BIT 6	13					
J8	ADR BIT 5	14					
J9	ADR BIT 5	14					
J10	SCE BIT 15	34					
J11	SCE BIT 15	34					
J12	SCE BIT 15	34					
J13	SCE BIT 15	34					
J14	SCE BIT 14	35					
J15	SCE BIT 14	35					
J16	SCE BIT 14	35					
J17	SCE BIT 14	35					
J18	SCE BIT 13	36					
J19	SCE BIT 13	36					
J20	SCE BIT 13	36					
J21	SCE BIT 13	36	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	SCE BIT 12	37	1.6	0.0	0.5	V	1-4
J23	SCE BIT 12	37					
J24	SCE BIT 12	37					
J25	SCE BIT 12	37					
J26	SCE BIT 11	38					
J27	SCE BIT 11	38					
J28	SCE BIT 11	38					
J29	SCE BIT 11	38					
J30	SCE BIT 10	39					
J31	SCE BIT 10	39					
J32	SCE BIT 10	39					
J33	SCE BIT 10	39					
J34	SCE BIT 9	41					
J35	SCE BIT 9	41					
J36	SCE BIT 9	41					
J37	SCE BIT 9	41					
J38	SCE BIT 8	42					
J39	SCE BIT 8	42					
J40	SCE BIT 8	42					
J41	SCE BIT 8	42					
J42	SCE BIT 7	43	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J43	SCE BIT 7	43	1.6	0.0	0.5	V	1-4
J44	SCE BIT 7	43					
J45	SCE BIT 7	43					
J46	SCE BIT 6	44					
J47	SCE BIT 6	44					
J48	SCE BIT 6	44					
J49	SCE BIT 6	44					
J50	SCE BIT 5	45					
J51	SCE BIT 5	45					
J52	SCE BIT 5	45					
J53	SCE BIT 5	45					
J54	SCE BIT 4	46					
J55	SCE BIT 4	46					
J56	SCE BIT 4	46					
J57	SCE BIT 4	46					
J58	SCE BIT 3	47					
J59	SCE BIT 3	47					
J60	SCE BIT 3	47					
J61	SCE BIT 3	47					
J62	SCE BIT 2	48					
J63	SCE BIT 2	48	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J64	SCE BIT 2	48	1.6	0.0	0.5	V	1-4
J65	SCE BIT 2	48					
J66	SCE BIT 1	49					
J67	SCE BIT 1	49					
J68	SCE BIT 1	49					
J69	SCE BIT 1	49					
J70	SCE BIT 0	50					
J71	SCE BIT 0	50					
J72	SCE BIT 0	50					
J73	SCE BIT 0	50					
J74	ADR BIT 4	52					
J75	ADR BIT 4	52					
J76	ADR BIT 3	53					
J77	ADR BIT 3	53					
J78	ADR BIT 2	54					
J79	ADR BIT 2	54					
J80	ADR BIT 1	55					
J81	ADR BIT 1	55					
J82	ADR BIT 0	56					
J83	ADR BIT 0	56	1.6	0.0	0.5	V	1-4
J84	EN TAB UP	58	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J85	DESTN BIT 15	59	1.6	0.0	0.5	V	1-4
J86	DESTN BIT 14	61					
J87	DESTN BIT 13	62					
J88	DESTN BIT 12	63					
J89	DESTN BIT 11	64					
J90	DESTN BIT 10	65					
J91	DESTN BIT 9	66					
J92	DESTN BIT 8	67					
J93	DESTN BIT 7	68					
J94	DESTN BIT 6	69					
J95	DESTN BIT 5	70					
J96	DESTN BIT 4	71					
J97	DESTN BIT 3	72					
J98	DESTN BIT 2	73					
J99	DESTN BIT 1	74					
J100	DESTN BIT 0	76	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J100
Power supply current limit	3.75 A	NA	J1-J100
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1-J3 J4-J83 J84 J85-J100
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J100 J1-J100
Output voltage measurement	NA	± 25 mV	J1-J100

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	SCE BIT 15	34	2.4	-0.08	0.08	mA	1-3
K2	SCE BIT 14	35					
K3	SCE BIT 13	36					
K4	SCE BIT 12	37					
K5	SCE BIT 11	38					
K6	SCE BIT 10	39					
K7	SCE BIT 9	41					
K8	SCE BIT 8	42					
K9	SCE BIT 7	43					
K10	SCE BIT 6	44					
K11	SCE BIT 5	45					
K12	SCE BIT 4	46					
K13	SCE BIT 3	47					
K14	SCE BIT 2	48					
K15	SCE BIT 1	49					
K16	SCE BIT 0	50	2.4	-0.08	0.08	mA	1-3
Low level							
K17	SCE BIT 15	34	0.4	-0.08	0.08	mA	1-3
K18	SCE BIT 14	35					
K19	SCE BIT 13	36	0.4	-0.08	0.08	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K20	SCE BIT 12	37	0.4	-0.08	0.08	mA	1-3
K21	SCE BIT 11	38					
K22	SCE BIT 10	39					
K23	SCE BIT 9	41					
K24	SCE BIT 8	42					
K25	SCE BIT 7	43					
K26	SCE BIT 6	44					
K27	SCE BIT 5	45					
K28	SCE BIT 4	46					
K29	SCE BIT 3	47					
K30	SCE BIT 2	48					
K31	SCE BIT 1	49					
K32	SCE BIT 0	50	0.4	-0.08	0.08	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K32
Power supply current limit	3.75 A	NA	K1-K32
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K16 K17-K32
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K32 K1-K32
Output current measurement	NA	± 8 uA	K1-K32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	IR BIT 4	21	-18.0	-1.2	0.0	V	1-3
L2	IR BIT 5	22					
L3	IR BIT 6	23					
L4	IR BIT 7	24					
L5	IR BIT 8	26					
L6	PH S/D CLK	27					
L7	SCE A EN	28					
L8	IR BIT 40	29					
L9	IR BIT 23	30					
L10	IR BIT 22	32					
L11	IR BIT 21	33	-18.0	-1.2	0.0	V	1-3
L12	EN ADR UP	57	-18.0	-1.5	0.0	V	1-3
L13	EN M ADR BIT	77	-18.0	-1.2	0.0	V	1-3
L14	MACLR SYNC	78					
L15	IR BIT 11	79					
L16	IR BIT 10	80					
L17	IR BIT 9	82					
L18	IR BIT 20	83					
L19	SCE B EN	84					
L20	IR BIT 16	85	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	IR BIT 19	86	-18.0	-1.2	0.0	V	1-3
L22	IR BIT 18	87					
L23	IR BIT 17	88					
L24	CONT FRMT	89					
L25	IR BIT 15	91					
L26	IR BIT 14	92					
L27	IR BIT 13	93					
L28	IR BIT 12	94					
L29	IR BIT 27	96					
L30	IR BIT 24	97					
L31	IR BIT 28	98					
L32	IR BIT 26	99					
L33	IR BIT 25	100	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L33
Power supply current limit	NA	NA	L1-L33
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L33
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L33 L1-L33
Input voltage measurement	 NA NA NA	 ± 60 mV ± 75 mV ± 60 mV	 L1-L11 L12 L13-L33

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1,31, 51,81	9.0	16.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 0.40 uF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test ciruit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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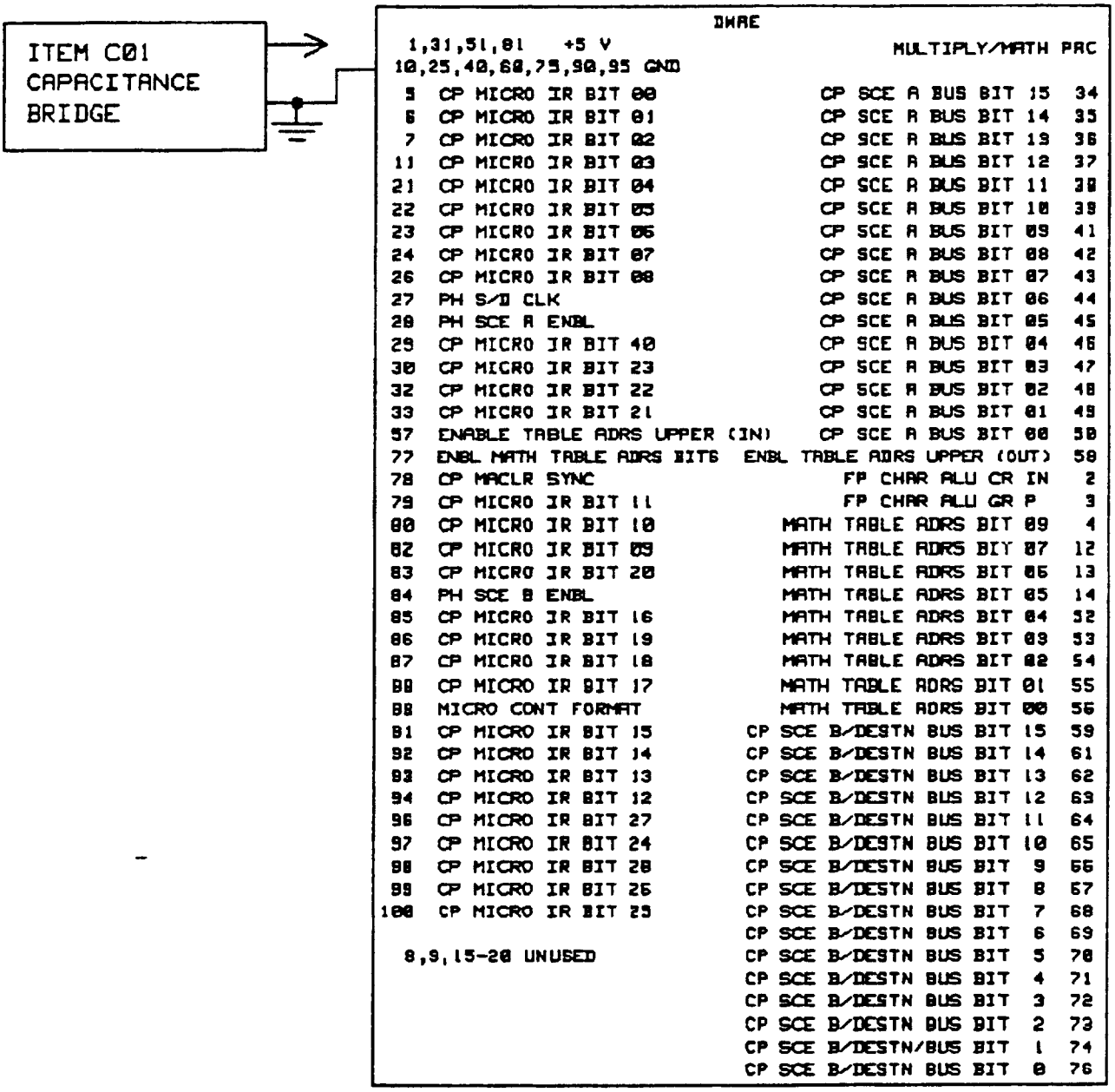


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	600	2500	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.75 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 95 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (300 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (300 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	3.75 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

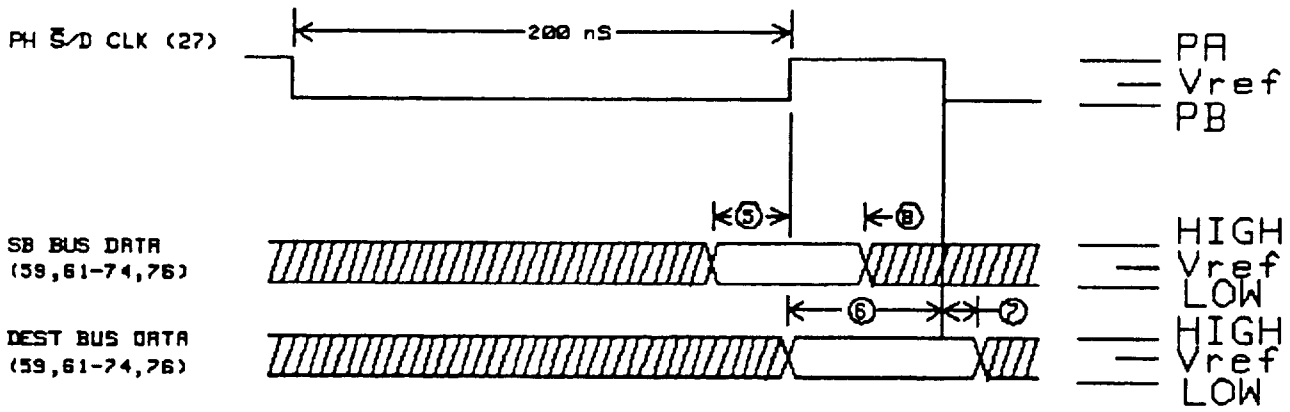
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. (5) DESTN BITS to PH S/D CLK, setup time maximum of 29 nS.
2. (8) DESTN BITS to PH S/D CLK, setup time maximum of 46 nS.
3. (6) PH S/D CLK to DESTN BITS, hold time minimum of 13 nS.
4. (7) PH S/D CLK to DESTN BITS, hold time minimum of 2 nS.

FIGURE 11. Input timing parameters for table XV.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (300 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (300 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (300 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (300 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	3.75 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

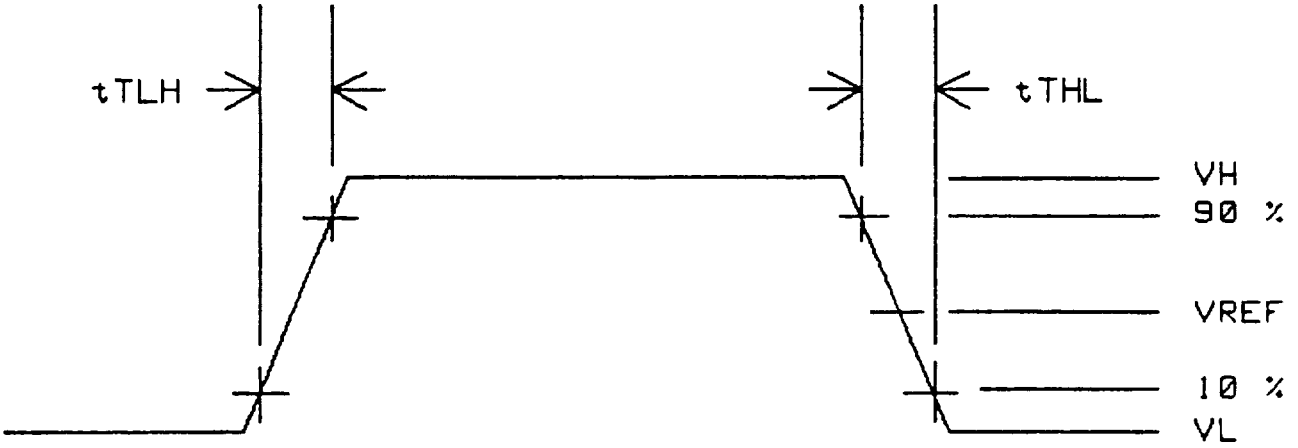
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	PH S/D CLK to SCE BIT 15	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 14	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 13	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 12	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 11	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 10	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 09	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 08	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 07	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 06	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 05	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 04	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 03	-	-	-	107	107	107	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	PH S/D CLK to SCE BIT 02	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 01	-	-	-	107	107	107	nS	12
1	tpd	PH S/D CLK to SCE BIT 00	-	-	-	107	107	107	nS	12
2	tpd	PH S/D CLK to DESTN BIT 15	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 14	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 13	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 12	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 11	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 10	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 09	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 08	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 07	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 06	-	-	-	95	95	95	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	PH S/D CLK to DESTN BIT 05	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 04	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 03	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 02	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 01	-	-	-	95	95	95	nS	12
2	tpd	PH S/D CLK to DESTN BIT 00	-	-	-	95	95	95	nS	12
3	tpd	PH S/D CLK to DESTN BIT 15	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 14	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 13	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 12	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 11	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 10	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 09	-	-	-	146	146	146	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	PH S/D CLK to DESTN BIT 08	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 07	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 06	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 05	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 04	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 03	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 02	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 01	-	-	-	146	146	146	nS	12
3	tpd	PH S/D CLK to DESTN BIT 00	-	-	-	146	146	146	nS	12
4	tpd	PH S/D CLK to DESTN BIT 15	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 14	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 13	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 12	-	-	-	182	182	182	nS	12

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Table XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	PH S/D CLK to DESTN BIT 11	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 10	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 09	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 08	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 07	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 06	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 05	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 04	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 03	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 02	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 01	-	-	-	182	182	182	nS	12
4	tpd	PH S/D CLK to DESTN BIT 00	-	-	-	182	182	182	nS	12
5	tsu	DESTN BIT 15 to PH S/D CLK	29	29	29	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	DESTN BIT 14 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 13 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 12 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 11 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 10 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 09 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 08 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 07 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 06 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 05 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 04 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 03 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 02 to PH S/D CLK	29	29	29	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tsu	DESTN BIT 01 to PH S/D CLK	29	29	29	-	-	-	nS	12
5	tsu	DESTN BIT 00 to PH S/D CLK	29	29	29	-	-	-	nS	12
6	tsu	DESTN BIT 15 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 14 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 13 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 12 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 11 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 10 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 09 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 08 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 07 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 06 to PH S/D CLK	46	46	46	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tsu	DESTN BIT 05 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 04 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 03 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 02 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 01 to PH S/D CLK	46	46	46	-	-	-	nS	12
6	tsu	DESTN BIT 00 to PH S/D CLK	46	46	46	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 15	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 14	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 13	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 12	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 11	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 10	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 09	13	13	13	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	th	PH S/D CLK to DESTN BIT 08	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 07	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 06	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 05	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 04	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 03	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 02	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 01	13	13	13	-	-	-	nS	12
7	th	PH S/D CLK to DESTN BIT 00	13	13	13	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 15	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 14	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 13	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 12	2	2	2	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Symbol	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	th	PH S/D CLK to DESTN BIT 11	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 10	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 09	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 08	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 07	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 06	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 05	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 04	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 03	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 02	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 01	2	2	2	-	-	-	nS	12
8	th	PH S/D CLK to DESTN BIT 00	2	2	2	-	-	-	nS	12

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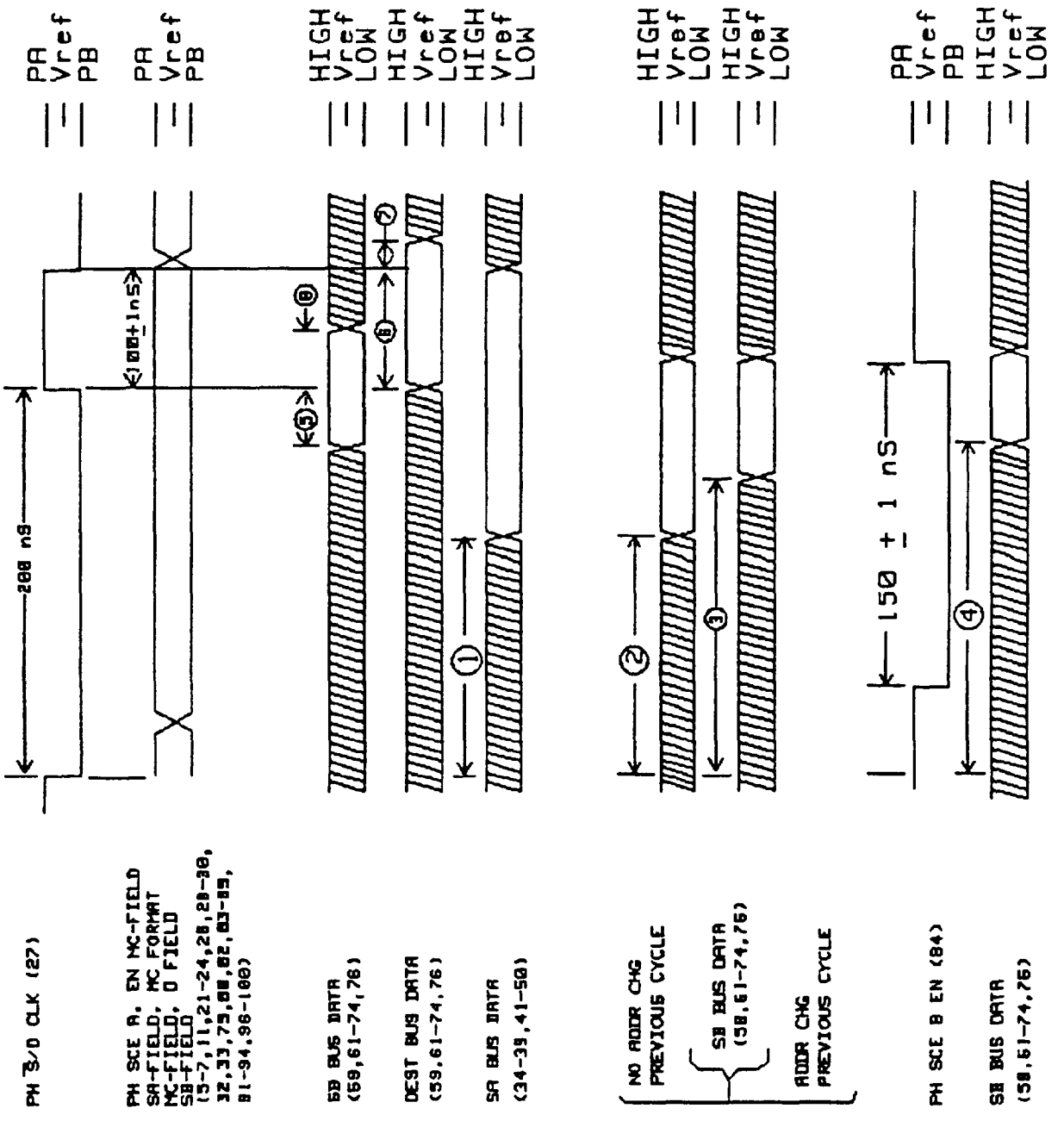


FIGURE 12. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	IR BIT 0	5	2	75	pF	1-4
R2	IR BIT 1	6				
R3	IR BIT 2	7	2	75	pF	1-4
R4	IR BIT 3	11	4	75	pF	1-4
R5	IR BIT 4	21	4	125	pF	1-4
R6	IR BIT 5	22	4	150	pF	1-4
R7	IR BIT 6	23				
R8	IR BIT 7	24	4	150	pF	1-4
R9	IR BIT 8	26	2	75	pF	1-4
R10	PH S/D CLK	27	18	228	pF	1-4
R11	SCE A EN	28	2	75	pF	1-4
R12	IR BIT 40	29				
R13	IR BIT 23	30				
R14	IR BIT 22	32				
R15	IR BIT 21	33				
R16	EN ADR UP	57				
R17	EN M ADR BIT	77	2	75	pF	1-4
R18	MACLR SYNC	78	8	125	pF	1-4
R19	IR BIT 11	79	4	75	pF	1-4
R20	IR BIT 10	80	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R21	IR BIT 9	82	2	75	pF	1-4
R22	IR BIT 20	83				
R23	SCE B EN	84				
R24	IR BIT 16	85				
R25	IR BIT 19	86				
R26	IR BIT 18	87				
R27	IR BIT 17	88				
R28	CONT FRMT	89				
R29	IR BIT 15	91				
R30	IR BIT 14	92				
R31	IR BIT 13	93				
R32	IR BIT 12	94				
R33	IR BIT 27	96				
R34	IR BIT 24	97				
R35	IR BIT 28	98				
R36	IR BIT 26	99				
R37	IR BIT 25	100	2	75	pF	1-4
Bidirectionals						
R38	ADR BIT 8	4	2	125	pF	1-4
R39	ADR BIT 7	12	2	175	pF	1-4
R40	ADR BIT 6	13	2	175	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	ADR BIT 5	14	2	175	pF	1-4
Outputs						
R42	SCE BIT 15	34	8	125	pF	1-4
R43	SCE BIT 14	35				
R44	SCE BIT 13	36				
R45	SCE BIT 12	37				
R46	SCE BIT 11	38				
R47	SCE BIT 10	39	8	125	pF	1-4
R48	SCE BIT 9	41	8	150	pF	1-4
R49	SCE BIT 8	42				
R50	SCE BIT 7	43				
R51	SCE BIT 6	44				
R52	SCE BIT 5	45				
R53	SCE BIT 4	46	8	150	pF	1-4
R54	SCE BIT 3	47	8	175	pF	1-4
R55	SCE BIT 2	48	8	175	pF	1-4
R56	SCE BIT 1	49	8	150	pF	1-4
R57	SCE BIT 0	50	8	175	pF	1-4
Bidirectionals						
R58	ADR BIT 4	52	8	150	pF	1-4
R59	ADR BIT 3	53	8	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R60	ADR BIT 2	54	8	125	pF	1-4
R61	ADR BIT 1	55	8	125	pF	1-4
R62	ADR BIT 0	56	8	150	pF	1-4
R63	DESTN BIT 15	59	4	75	pF	1-4
R64	DESTN BIT 14	61				
R65	DESTN BIT 13	62				
R66	DESTN BIT 12	63				
R67	DESTN BIT 11	64				
R68	DESTN BIT 10	65				
R69	DESTN BIT 9	66				
R70	DESTN BIT 8	67				
R71	DESTN BIT 7	68				
R72	DESTN BIT 6	69				
R73	DESTN BIT 5	70	4	75	pF	1-4
R74	DESTN BIT 4	71	4	100	pF	1-4
R75	DESTN BIT 3	72	4	100	pF	1-4
R76	DESTN BIT 2	73	4	75	pF	1-4
R77	DESTN BIT 1	74				
R78	DESTN BIT 0	76	4	75	pF	1-4
R79	ALU CR IN	2	2	125	pF	1-4
R80	ALU GR P	3	2	125	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R80
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R80
Capacitance measurement	NA	± 3.65 pF	R1-R3
	NA	± 3.55 pF	R4
	NA	± 6.05 pF	R5
	NA	± 7.3 pF	R6-R8
	NA	± 3.65 pF	R9
	NA	± 10.5 pF	R10
	NA	± 3.65 pF	R11-R17
	NA	± 5.85 pF	R18
	NA	± 3.55 pF	R19,R20
	NA	± 3.65 pF	R21-R37
	NA	± 6.15 pF	R38
	NA	± 8.65 pF	R39-R41
	NA	± 5.85 pF	R42-R47
	NA	± 7.1 pF	R48-R53
	NA	± 8.35 pF	R54,R55
	NA	± 7.1 pF	R56
	NA	± 8.35 pF	R57
	NA	± 7.1 pF	R58
	NA	± 5.85 pF	R59-R61
	NA	± 7.1 pF	R62
	NA	± 3.55 pF	R63-R73
	NA	± 4.8 pF	R74,R75
	NA	± 3.55 pF	R76-R78
	NA	± 6.15 pF	R79,R80

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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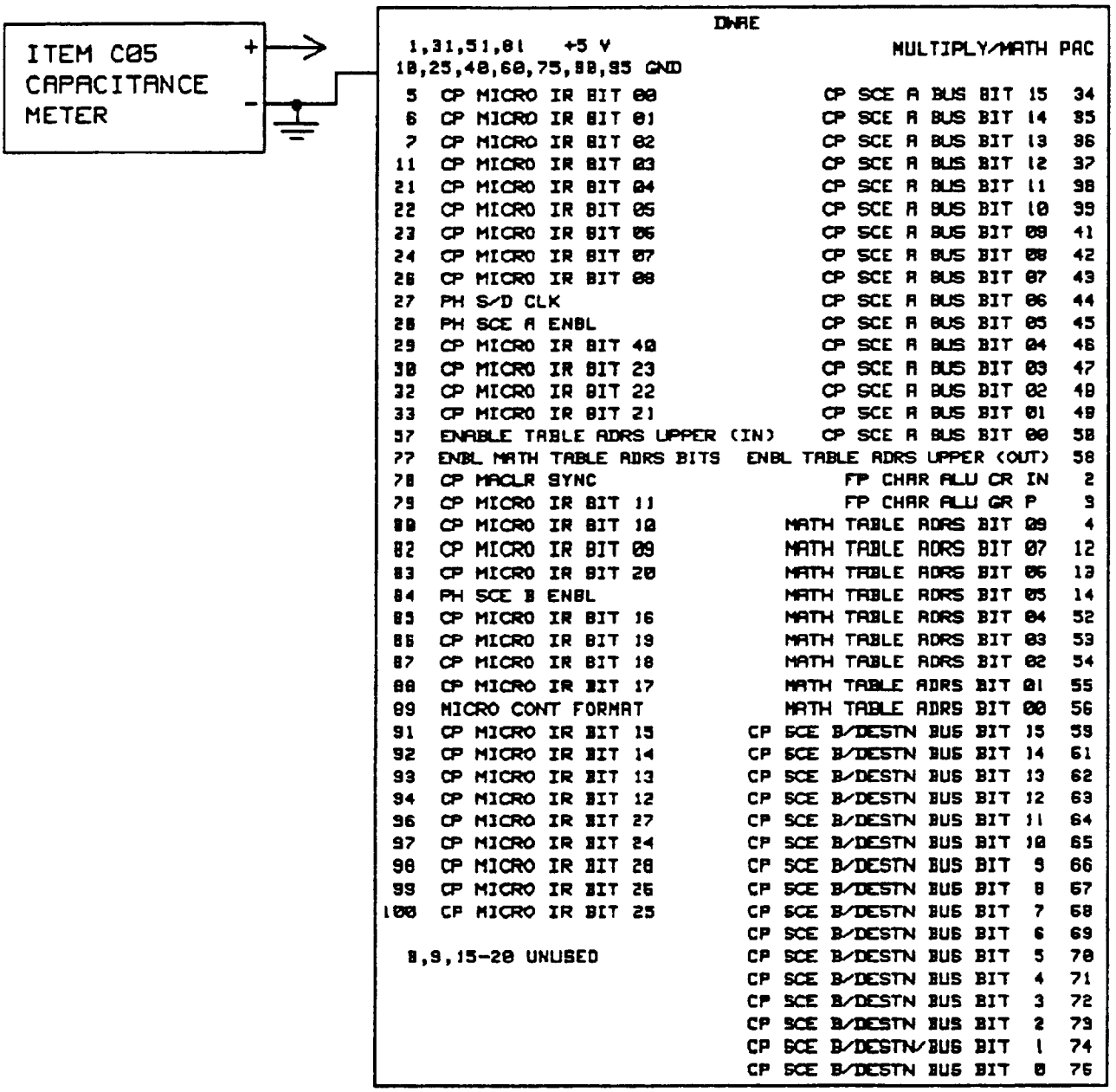


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T	PSTA	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	3.75 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

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NOTES: - Continued.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', connected to Vcc at the point of shorting.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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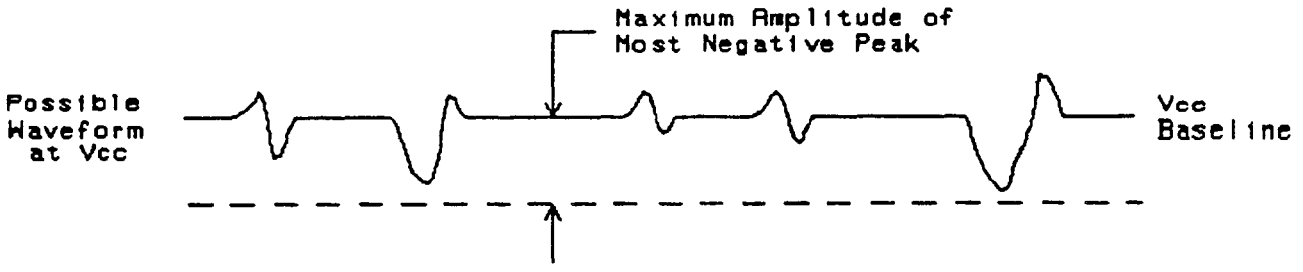
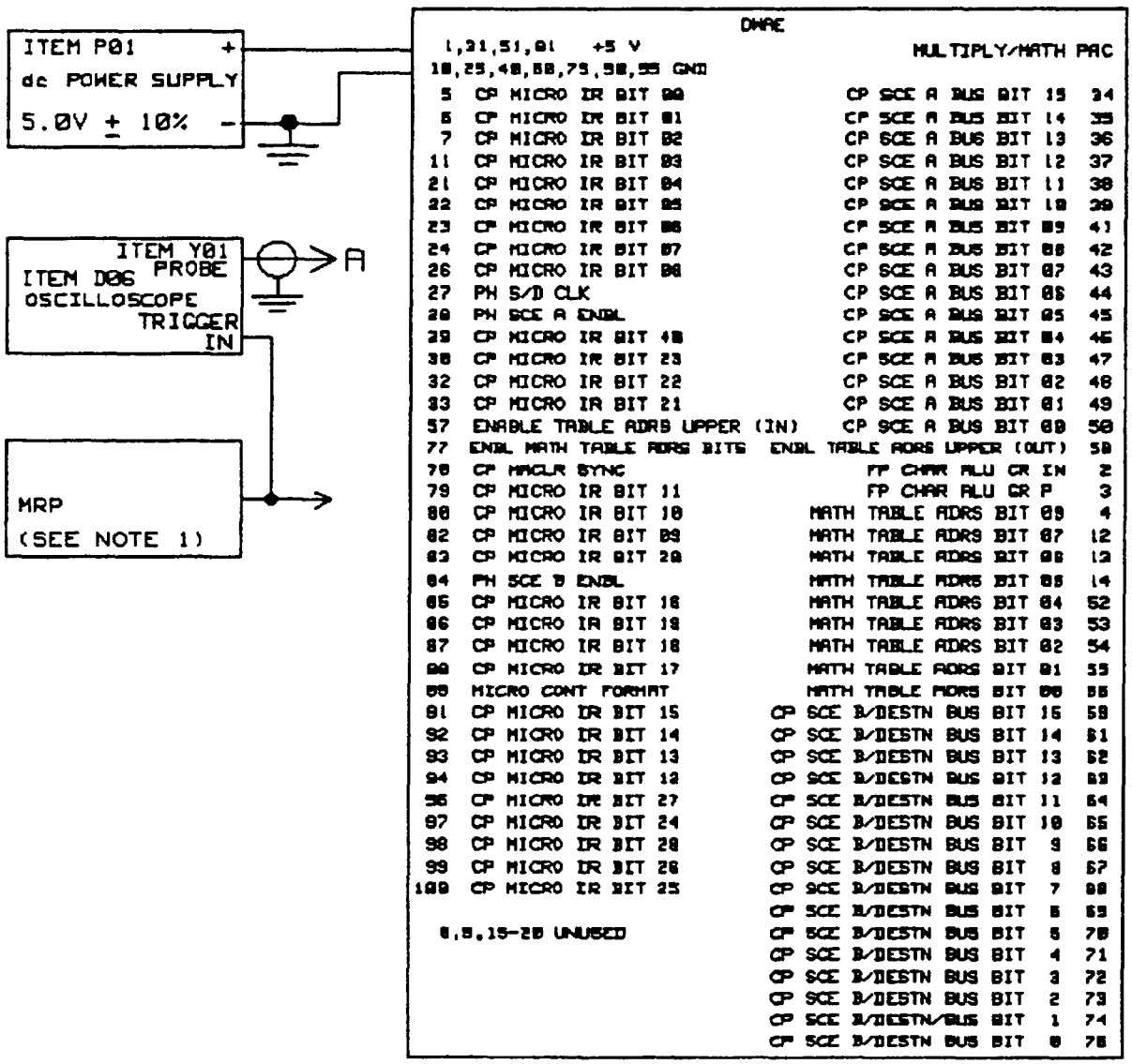


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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